Part VIII

Commodity, Environmental, and Digital Markets

In these markets the standard assumptions bend: in Chapter 36 because the underlying is a physical thing or a policy, in Chapter 37 because the underlying is well behaved for carry and the infrastructure around the contract is missing. Storage costs, delivery constraints, weather, emissions allowances, and a perpetual contract with no expiration each break something the earlier chapters took for granted. Both chapters can be read on their own; where they draw on Parts VI and VII, the point is restated where it is used.

36 Energy, Commodity, and Environmental Derivatives
carry with storage, spreads, weather, and carbon

37 Crypto Derivatives
perpetuals, funding rates, and derivatives without a central counterparty

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Chapter 36

Energy, Commodity, and Environmental Derivatives

Section 5.8 priced a commodity forward with storage costs and a convenience yield, and Section 5.14 left to this chapter the markets where the term structure carries the most information. What makes it carry information is a single asymmetry. In equity and rates, the forward curve is derived from the spot by arbitrage; a violation is a trade. In commodities the arbitrage runs one way only—you can buy and store, but you cannot short and un-store—so the curve carries information about physical conditions that no arbitrage forces into line.

That asymmetry generates everything in this chapter. The curve's shape means something, seasonality survives, volatility depends on maturity, and one commodity, electricity, breaks the framework entirely because it cannot be stored at all.

The chapter then follows the asymmetry to its limit. Emissions allowances are storable at no cost, so the carry relation binds tighter than for any physical commodity and its residual means something else; allowance options have a bimodal density and a smile that is not a smile; and weather and catastrophe contracts have an underlying that cannot be held, so the writer cannot hedge and the price is an expectation rather than a replication cost.

Sections 36.1 to 36.3 read the curve: what its shape says about inventories, why seasonality survives, and why volatility depends on maturity. Sections 36.4 to 36.8 take electricity and the contracts written on it, the spreads that turn one commodity into another, storage as an option, and the models these markets need; Sections 36.9 to 36.12 cover basis, the stack-and-roll hedge, financial participation, and the remaining markets. Sections 36.13 to 36.16 follow the asymmetry to its limit through emissions, weather, and catastrophe risk, and Section 36.17 sorts what is priced by arbitrage from what is priced by expectation. Section 36.18 says who trades these markets and why, and Section 36.19 how the chapter's results were obtained.

36.1 THE CURVE IS THE INFORMATION

Section 5.8's relation is

$$F(t,T) = S(t)\, e^{(r+u-y)(T-t)} \tag{36.1}$$

with u the storage cost and y the convenience yield, the benefit of holding the physical commodity rather than a claim on it. The convenience yield is not observed: it is the residual that makes the equation hold, and treating it as a parameter with an independent existence is the most common error in commodity finance. But the residual is informative. A convenience yield above $r + u$ (a market in backwardation) says physical inventory is scarce and someone will pay to hold it now. A lower one (contango) says inventory is plentiful and the curve is compensating storage. Two consequences follow that have no analog in financial markets.

For a storable commodity, the upper bound on the curve is enforced and the lower bound is not. If F rises above full carry, $S(t)\, e^{(r+u)(T-t)}$, anyone can buy spot, store, and sell forward. If F falls below it, the arbitrage requires borrowing the physical commodity, and for most commodities that market is thin or nonexistent. So contango is capped by storage economics

and backwardation is not capped at all, which is why the deepest backwardations occur in genuine shortages and can be extreme.

Storage capacity is also finite. When it fills, the cap disappears: the spot and front-month prices can fall any distance below the later contracts, because nobody can buy the commodity and store it. The most-quoted case is the April 2020 crude oil episode, when the front-month contract settled below zero, with later months still positive, because holders faced physical delivery with no storage available. That was not a market failure but the arbitrage bound behaving as it must when the constraint binds, and the clearest modern demonstration that a commodity futures contract is a claim on a physical thing.

Example 36.1 Reading inventory off the curve

Crude spot \$78.00, financing 4.5%, storage 6% of value per year. Inverting (36.1),

$$y = r + u - \frac{1}{T}\ln\frac{F}{S}$$

	Backwardated		Contango	
Months	F	**Implied** y	F	**Implied** y
1	77.40	19.77%	78.35	5.13%
3	76.30	19.31%	79.10	4.90%
6	75.00	18.34%	80.20	4.94%
12	73.20	16.85%	82.00	5.50%
24	71.50	14.85%	84.50	6.50%

In backwardation the implied yield far exceeds $r + u$, which is the market paying nearly 20% a year for the privilege of holding a physical barrel now. The term structure of y is itself a forecast: it declines with maturity in the backwardated case, which says the shortage is expected to resolve, where a flat implied-yield curve would say the tightness is expected to persist.

A negative implied convenience yield is the diagnostic for full storage. It means the curve is steeper than full carry, which cannot be arbitraged away because there is nowhere to put the commodity. That steepness is the precondition for the April 2020 episode, but not the episode itself: (36.1) makes F a positive multiple of S, so no convenience yield, however negative, can take a futures price below zero. What went negative was the spot value of a barrel at the delivery point. Holders of the expiring contract had to take delivery with no tank to put it in, so owning a barrel had become a disposal cost, and on April 20, 2020 the May contract settled at −\$37.63 while June, a month later, settled at +\$20.43.

36.2 SEASONALITY

Financial forward curves are smooth; commodity curves are not. Natural gas, power, and agricultural curves have a stable annual shape driven by consumption:

Month	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Gas forward (\$/MMBtu)	2.85	2.88	2.95	3.05	3.08	3.02	3.10	3.45	3.80	3.95	3.85	3.30

The winter premium is not a forecast that prices will rise. It is the cost of the storage cycle: gas injected in summer and withdrawn in winter must earn the storage cost, the financing, and a return on the facility. Two practical consequences follow.

A "roll yield" computed across a seasonal boundary is meaningless. Rolling from October into November in gas is not a term-structure trade; it is a seasonal step. Section 5.11's roll analysis applies within a season and not across one.

Seasonality must also be removed before a volatility is estimated: a series that rises every winter moves partly on a predictable schedule, and counting that movement as risk overstates

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the volatility. This is the seasonal adjustment Section 33.10 makes to an inflation curve, with a larger amplitude.

36.3 THE SAMUELSON MATURITY EFFECT

The **Samuelson maturity effect**, named for Paul Samuelson, who predicted it in 1965, is the tendency of a futures contract's volatility to rise as its expiration approaches. Seen across the curve on any one day, it means near contracts are more volatile than far ones. The reason is mean reversion in the spot.

Example 36.2 Volatility falls with maturity

If the spot mean-reverts at rate a, a shock decays before a distant contract expires, so the futures volatility is

$$\sigma_F(T) = \sigma_S\, e^{-a(T-t)}, \tag{36.2}$$

with σ_S the volatility of the spot price and $T - t$ the contract's time to expiration. [TN36] derives it from a mean-reverting log spot price. The decay is the one Section 31.6 found for Hull–White forward rates, and for the same reason: mean reversion damps a shock before the distant date arrives. The quantities differ, though. There the short rate mean-reverts and the result is a basis-point volatility of a forward rate; here the logarithm of the spot price mean-reverts and the result is a percentage volatility of a futures price. At a 60% spot volatility:

Months out	$a = 0.5$	$a = 1.5$	$a = 3.0$
1	57.6%	52.9%	46.7%
3	52.9%	41.2%	28.3%
6	46.7%	28.3%	13.4%
12	36.4%	**13.4%**	3.0%
24	22.1%	3.0%	0.1%

At $a = 1.5$ the one-month contract carries 52.9% volatility and the twelve-month 13.4%, a factor of 3.96. Three things follow.

Options on different contract months are not comparable at one volatility, because each month is a different underlying with its own volatility. A 40% implied volatility is unremarkable on the front month and extreme on the one-year contract, and a volatility surface for a commodity needs a maturity dimension that means something different from an equity surface's.

Hedge ratios across maturities are not one-for-one. A one-for-one hedge of a two-year exposure with front-month futures is too large by the ratio of their volatilities in the table, about 18 at $a = 1.5$, which is the commodity version of the duration mismatch that Section 6.7's duration-based hedge corrects.

Storage works against the effect. It links adjacent contracts, which *raises* the correlation between them and slows the mean reversion that *lowers* the far contract's volatility. A commodity with cheap, ample storage has a flat volatility term structure and a highly correlated curve; one with costly or full storage has a steep one and a curve whose near and far ends can move independently. The volatility term structure is therefore a direct read on storage economics.

Figure 36.1 shows both effects: futures volatility falling with maturity, and a gas curve whose shape is set by the calendar rather than by the price level.

36.4 ELECTRICITY

Power cannot be stored, and that single fact removes most of this book's machinery. Equation (36.1) does not hold, because there is no cash-and-carry: you cannot buy power today and deliver it tomorrow. So the forward is not linked to the spot by arbitrage at all: it is an expectation plus a risk premium, and nothing enforces it.

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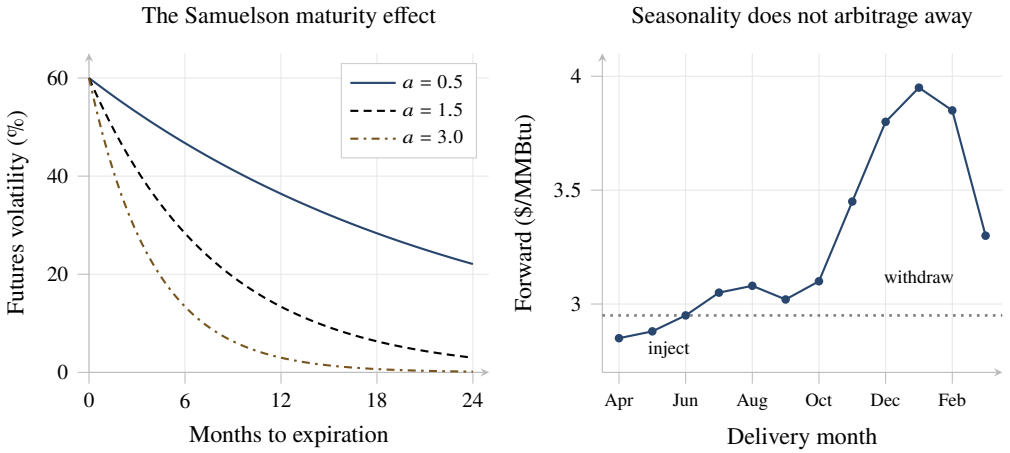


Figure 36.1: Left: equation (36.2) at $\sigma_S = 60\%$. Right: Section 36.2's gas curve.

The price process is correspondingly unusual. Supply is a stack of generators ordered by marginal cost, and the price is set by the last unit needed. When demand approaches capacity the stack becomes near-vertical, so a small demand increase produces an enormous price move—and then reverts within hours when the weather changes.

A simple model captures this. On an ordinary day the price is lognormal; on a small fraction p of days demand reaches the steep end of the stack, and the day's price is multiplied by a random spike factor J:

$$P = m\, e^{\sigma Z} J^{B}, \qquad Z \sim N(0,1), \quad B \sim \text{Bernoulli}(p), \quad \ln J \sim N\big(\mu_J, s_J^2\big), \tag{36.3}$$

with Z, B and J independent. Here m is the median ordinary-day price (half of the days without a spike settle below it, and their mean is the slightly higher $m\, e^{\sigma^2/2}$), σ is the standard deviation of its logarithm, and $E[J] = e^{\mu_J + s_J^2/2}$ the average size of a spike. Call this the **spike model**. Because J is lognormal, the price on a spike day is lognormal too, so the price is a mixture of two lognormals: with probability $1-p$ its logarithm is normal with mean $\ln m$ and variance σ^2, and with probability p normal with mean $\ln m + \mu_J$ and variance $\sigma^2 + s_J^2$. Its mean is

$$E[P] = m\, e^{\sigma^2/2}\big(1 - p + p\, E[J]\big),$$

and its distribution function is the probability-weighted sum of the two lognormals' distribution functions. The q-th percentile x_q therefore solves

$$(1-p)\, N\!\left(\frac{\ln(x_q/m)}{\sigma}\right) + p\, N\!\left(\frac{\ln(x_q/m) - \mu_J}{\sqrt{\sigma^2 + s_J^2}}\right) = q,$$

which has no closed-form solution but has exactly one root, because the left side rises steadily from 0 to 1 as x_q increases, so bisection finds it.

A contract on the day's price is valued the same way, by conditioning on whether the day spikes. A daily cap struck at K, paying $\max(P-K, 0)$, is worth

$$c(K) = (1-p)\, c^{\mathrm{B}}\big(F_0, K, \sigma\big) + p\, c^{\mathrm{B}}\Big(E[J]\, F_0,\ K,\ \sqrt{\sigma^2 + s_J^2}\Big), \qquad F_0 = m\, e^{\sigma^2/2}, \tag{36.4}$$

stated at the delivery date, before discounting, where

$$c^{\mathrm{B}}(F, K, v) = F\, N(d_1) - K\, N(d_1 - v), \qquad d_1 = \frac{\ln(F/K) + v^2/2}{v}$$

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is Black's formula from Chapter 16 written with the total volatility v in place of $\sigma\sqrt{T}$. Each term is one regime: an ordinary day has mean F_0 and log standard deviation σ; a spike day has a mean $E[J]$ times larger, and the spike factor's own uncertainty adds to the variance. A floor is valued the same way with Black's put, and a forward on the day's price is worth $E[P]$. So every daily figure the model produces is exact.

The model has no time dimension. It describes the distribution of one day's price, with each day drawn independently and a spike lasting exactly one day: the "reverts within hours" above, taken to its limit. That is all a daily cap needs, because its payoff depends on one day's price and a strip of daily caps is worth the sum of its days. A payoff that depends on several days (a cap on a monthly average, a swing contract, a hedge carried through a spike) needs the path as well, and there the standard choice is a mean-reverting log price with jumps that decay within days, so that spikes arrive in clustered episodes rather than as independent days.

Example 36.3 Why a lognormal model misprices a price cap

Take the spike model (36.3) with an ordinary-day median $m = \$43$ and $\sigma = 30\%$. Spikes occur on 2.4% of delivery days ($p = 2.4\%$), about nine days in a 365-day year, with a spike factor of mean $E[J] = 5$ and $s_J = 60\%$. Against it, fit a single lognormal with the same mean (\$49.30) and the same standard deviation (\$40.42) and compare.

	Spike model	Matched lognormal
Mean	\$49.30	\$49.30
Median	\$43.39	\$38.12
99th percentile	\$206.81	\$202.15
99.9th percentile	\$573.80	\$349.60

Now price daily caps with (36.4). At a \$100 strike, $F_0 = 43\, e^{0.30^2/2} = \$44.98$, so an ordinary day is worth $c^{\mathrm{B}}(44.98, 100, 0.30) = 0.0238$; a spike day, with mean \$224.90 and total volatility $\sqrt{0.09 + 0.36} = 67.1\%$, is worth $c^{\mathrm{B}}(224.90, 100, 0.671) = 130.24$. Weighting by the probabilities,

$$c(100) = 0.976 \times 0.0238 + 0.024 \times 130.24 = 0.0232 + 3.1257 = 3.1490,$$

so 99% of the cap's value comes from the nine or so spike days a year. The matched lognormal is a single application of Black's formula, at the forward \$49.30 and the 71.7% volatility that reproduces the standard deviation of \$40.42. Across strikes:

Strike	Spike model	Lognormal	Ratio
\$ 60	5.3300	10.4989	0.51
\$100	3.1490	4.1344	0.76
\$200	1.6578	0.6518	**2.54**
\$400	0.5056	0.0483	**10.48**
\$800	0.0739	0.00158	47

Matched on mean and on volatility, the two models disagree by a factor of ten at a \$400 strike—and they disagree in opposite directions at the two ends.

The lognormal spends its variance on everyday fluctuation and the real distribution spends it on nine or so days a year. So the lognormal over-prices near-the-money caps, because it makes moderate moves too likely, and drastically under-prices far ones.

A utility buying a price cap is buying precisely that difference, and it is one of the largest model risks in energy trading. No amount of calibrating a lognormal to at-the-money options fixes it. The lognormal already over-prices the cap near the money, so fitting that price lowers its volatility, and a lower volatility cheapens the far caps further.

Two further features of electricity have no financial-market analog.

Prices can be negative, and routinely are in systems with large inflexible or subsidized

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generation, because a generator may prefer to pay to keep running rather than shut down and restart. Any model requiring positivity, and any option formula written in log-space, cannot represent them, which is why power desks often work in normal rather than lognormal terms, as Section 29.8's rate markets did. Neither model in Example 36.3 can produce a negative price: both multiply positive factors, so the spike model repairs the upper tail and is silent about the lower. Its cap prices are unaffected, because a cap pays only in the upper tail. Where negative prices matter (a floor, or a generator with an obligation to keep running), the ordinary-day price has to be modeled additively, as a normal variable or a lognormal shifted below zero, with the spike added on top.

The commodity is defined by location and by hour. Power at one node and at another is a different good, connected by transmission that can bind. The basis between locations is itself traded, as the financial transmission rights of Table 36.1. It is close to zero while the line has spare capacity and jumps when the line binds, so it behaves less like a spread between two prices than like an option that pays only in the hours the line is congested.

36.5 ELECTRICITY CONTRACTS AND HOW THEY ARE PRICED

Because nothing links the forward to the spot, an electricity market is a family of contracts, each written on a different slice of the same price (a delivery period, a day, an hour, a location), and each priced by its own method. Table 36.1 sets them out.

Three rules organize the last column.

The forward curve is an input. Power cannot be carried, so (36.1) produces no forward price. A block forward is the expected average spot price over its delivery period under the risk-neutral measure, and nothing arbitrages it toward today's spot. Desks therefore take the curve from the market and calibrate every other model to it, which reverses Chapter 5's order: there the spot and the cost of carry gave the forward, and here the forwards are the data.

The settlement period picks the model. A contract settled on an average over a month depends on how the whole delivery period moves and is priced with a model of the forward curve. A contract settled on one day or one hour depends on spikes and is priced with a model of the spot. A block forward averages hundreds of hourly prices, so its volatility is a fraction of the spot's, and the Samuelson maturity effect lowers it further the farther away delivery is. An option on a block is therefore cheap next to a strip of daily caps at the same strike. The next example shows by how much, and why the answer turns on something Example 36.3 could leave out.

Flexibility is a chain of linked options. The clearest case is the **swing contract**, a supply agreement at a fixed price in which the buyer chooses each day how much to take, between a daily minimum and maximum, subject to a minimum and maximum over the whole contract and a penalty for taking less than the total minimum. A gas retailer holding one takes the maximum on cold days, when spot gas costs more than the contract price, and the minimum on mild ones; what links the days is the total limit, because volume taken in December is not available in February. Tolling agreements and storage work the same way: the holder chooses how much, or when, on many dates, and each choice uses up capacity that a later one would have needed. Their value therefore depends on the whole price path rather than on one day's price, and it is found by dynamic programming over the price and the remaining volume, or by Chapter 20's least-squares Monte Carlo when the state is too large for a lattice.

Example 36.4 A daily cap and a monthly cap are different instruments

The example shows that two caps on the same price at the same strike can differ in value by a factor of eight, and that the gap depends on how spikes are spread through time, which the spike model of

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Table 36.1: Electricity contracts, their users, and how each is priced.

Contract	What it pays	Main users	Priced by
Block futures and forwards	A fixed price for power over a month, quarter or year, baseload or peak hours; usually cash-settled on the average spot price over the delivery period	Generators sell; retailers and large consumers buy	The market: the curve is an input
Swaps	The same, over the counter: fixed against the average spot price at a hub or zone	Utilities, industrial consumers	As forwards
Options on block futures	Calls and puts on a block forward	Generators and retailers matching hedges to their load profile	Black's formula with the block's own volatility
Daily and hourly options; caps and floors	One option per delivery day or hour on that period's spot price	Retailers capping cost; owners of peaking plants	A spot model with spikes, Example 36.3
Spark and dark spread options; tolling	Power less fuel; a tolling agreement rents a plant's output	Plant owners and buyers of capacity	Exchange option, Example 36.5; dynamic programming for operating constraints
Swing and load-following contracts	A fixed price with volume at the holder's choice, within daily and total limits	Retail suppliers whose load follows the weather	Dynamic programming or least-squares Monte Carlo
Financial transmission rights	The price difference between two nodes, the congestion rent	Generators and load-serving utilities hedging location	Auction prices and network simulation; no replication
Power purchase agreements	A fixed price for a plant's output as produced; a virtual agreement settles the difference in cash	Corporate buyers of renewable power; wind and solar developers	A forward at the output-weighted price, plus shape, volume and basis risk

Example 36.3 does not specify. Take a 30-day month, a strike of \$60, and 1 MWh a day, with each day's price drawn from that example's spike model. A strip of daily caps pays $\sum_d \max(P_d - 60, 0)$; a cap on the monthly average pays $30\max(\bar{P} - 60, 0)$, with $\bar{P}$ the month's average price. Values are undiscounted.

The strip is worth $30 \times 5.3300 = \$159.90$, whatever the relation between days, because each of its terms depends on one day only. The average cap depends on the joint behavior of the days. The table varies that behavior while holding each day's distribution fixed: spikes arrive either on independent days or in episodes of three or five consecutive days, with the probability that a given day spikes kept at 2.4%. The average cap has no formula like (36.4), because its payoff depends on a sum of 30 prices, so it is valued by simulation: draw each day's ordinary price, whether the day spikes and its spike factor, average the month, and take the mean payoff over 400,000 simulated months for each row:

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Spikes arrive	S.d. of $\bar{P}$	Monthly-average cap	Daily strip
On single days	\$ 7.38	\$19.1	\$159.90
In three-day episodes	\$10.12	\$44.3	\$159.90
In five-day episodes	\$12.18	\$66.2	\$159.90

Averaging removes most of a spike. A spike day that multiplies the price by five raises a 30-day average by only about \$6, so with independent days the monthly cap is worth an eighth of the strip. That is why options on block futures trade at volatilities far below those of daily options in the same market: the block's payoff is an average, and an average does not see a single day.

Clustering raises the monthly cap and leaves the strip alone. Spikes in five-day episodes (a heat wave rather than a hot afternoon) more than triple the monthly cap, from \$19.1 to \$66.2. The dependence across days that Example 36.3 could ignore is the input that decides a monthly contract. A model calibrated to daily options says nothing about it, and a desk that prices monthly options as if days were independent will value them at less than half their worth if spikes come in three-day episodes, and at less than a third if they come in five-day ones.

36.6 SPARK, DARK, AND CRACK SPREADS

Many of the most important commodity derivatives are written not on a commodity but on the *margin between two*, because that is what an industrial owner actually earns.

Example 36.5 A power plant is an exchange option

A gas-fired generator earns $\max(P_e - \text{HR} \cdot P_g, 0)$ per MWh, where P_e is the power price per MWh, P_g the gas price per million British thermal units (MMBtu), and HR the **heat rate**, the MMBtu of gas the plant burns to produce one MWh. The margin $P_e - \text{HR} \cdot P_g$ is the **spark spread**. The plant runs when it is positive and shuts when it is not, so the plant *is* a strip of call options on the spark spread—and by Section 34.10 each is an exchange option: power against HR units of gas.

The example asks what one of those options is worth, and shows that the answer turns almost entirely on the correlation between power and gas. Take a peaking plant, one that runs only in the hours when power prices are high: power \$42.00/MWh, gas \$3.20/MMBtu, heat rate 12.5, six months, power volatility 55%, gas volatility 45%. Fuel cost is \$40.00/MWh, so the intrinsic spread is \$2.00. The option values below are forward values, undiscounted. *Effective vol* is the volatility in Margrabe's formula, $\sqrt{\sigma_e^2 + \sigma_g^2 - 2\rho\,\sigma_e\sigma_g}$, with σ_e and σ_g the two legs' volatilities.

ρ	Effective vol	Option value	Time value
0.9	24.4%	\$3.929	\$1.929
0.7	39.8%	\$5.658	\$3.658
0.5	50.7%	\$6.891	\$4.891
0.3	59.7%	\$7.900	\$5.900
0.0	71.1%	\$9.171	\$7.171

Power and gas are strongly correlated because gas frequently sets the marginal price of power, so the spread is far less volatile than either leg. If the true correlation is 0.9, modeling them as independent values the option on each megawatt-hour at \$9.171 instead of \$3.929, 133% too high. *High correlation makes a spread option cheap*.

This is among the most consequential correlation assumptions in energy finance, because it values the entire generation fleet: an unhedged generator's enterprise value is largely the value of a portfolio of spark-spread options, and Section 34.10's warning that correlation must come from somewhere applies here with a whole plant's value at stake.

The same structure recurs. A **dark spread** is the coal-fired version. A **crack spread** is the refiner's:

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Crude	\$78.00/bbl
Gasoline \$2.45/gal × 42	\$102.90/bbl
Heating oil \$2.60/gal × 42	\$109.20/bbl
3-2-1 crack = $(2 \times 102.90 + 109.20 - 3 \times 78.00)/3$	**\$27.00/bbl**

A refiner is long the spread and hedges by selling it: buying crude futures and selling product futures in the 3-2-1 ratio, three barrels of crude to two of gasoline and one of heating oil. The ratio is technological, not conventional: it approximates the yield of a typical US refinery, and a refiner with a different configuration hedging at 3-2-1 has a residual exposure to its own mix of products.

36.7 STORAGE IS AN OPTION

A storage facility is the right, not the obligation, to buy at one date and sell at another. Its value therefore splits the way an option's does, as Example 36.6 shows.

Example 36.6 Intrinsic and extrinsic

Using Section 36.2's curve, a facility can lock in a spread today: inject in the five cheapest months (April through September, skipping August, averaging \$2.95) and withdraw in the five most expensive (November through March, averaging \$3.67). The intrinsic value is \$0.72 per MMBtu per cycle, and it can be hedged immediately with calendar spreads.

The extrinsic value is the right to change one's mind: to withdraw early if winter arrives early and cold, to re-inject if the spread widens. It is the time value of a strip of calendar-spread options, and it depends on volatility and on how correlated the injection and withdrawal months are:

Volatility	$\rho = 0.90$	$\rho = 0.75$	$\rho = 0.50$
30%	0.2%	2.2%	7.1%
45%	1.7%	8.3%	17.4%
60%	5.1%	15.7%	27.0%
80%	11.0%	25.0%	**37.4%**

The entries are extrinsic value as a share of the total, with the strip valued as one exchange option (Margrabe's formula) on the \$3.67 withdrawal average against the \$2.95 injection average over six months, both at the volatility shown and correlation ρ between them. At plausible parameters the extrinsic share is 10–30%, and it is the part a valuation that locks in only today's spread does not capture.

This explains a market structure fact. Storage capacity is often leased by trading firms rather than operated by the utility that built it, and the reason is not that traders are better at running compressors. A utility that leases out its capacity for about the intrinsic spread has sold the extrinsic value to the lessee, because capturing it requires continuous re-optimization, which is an option-hedging operation, not a physical one.

The same decomposition prices every physical asset with flexibility: a pipeline is a locational spread option, a swing contract on gas is a strip of daily options linked by its volume limits, and a liquefied natural gas (LNG) cargo with destination flexibility is an option on the basis between two markets.

36.8 MODELING

Four features have to be accommodated and no single model does all of them well.

Mean reversion generates Section 36.3's volatility term structure and is the minimum requirement.

Seasonality is handled deterministically (a seasonal function added to the log price) rather than stochastically, because it is predictable and modeling it as risk overstates volatility.

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Two factors, at least, are needed for anything spread-related. A one-factor model makes all contract months perfectly correlated, which, once the model is fitted to a realistic curve, prices a calendar spread option at nearly zero. This is Section 31.12's defect in another market, and the fix is the same: a short-term mean-reverting factor plus a long-term factor with a random walk, which is the standard two-factor commodity model and produces both a decaying volatility term structure and imperfect correlation along the curve. Example 36.7 shows how much the second factor is worth.

Example 36.7 Why a calendar spread needs two factors

The example shows that a one-factor model fitted to the same market as a two-factor model can value a calendar spread option at about a twelfth of the two-factor price. Let the market follow the two-factor model

$$\ln S = \chi + \xi, \qquad d\chi = -\kappa\chi\, dt + \sigma_\chi\, dW_\chi, \qquad d\xi = \mu\, dt + \sigma_\xi\, dW_\xi,$$

with independent shocks, a short-term factor at $\sigma_\chi = 40\%$ and $\kappa = 2$, and a long-term factor at $\sigma_\xi = 20\%$. The argument of [TN36], applied to each factor, gives the futures volatility at time to delivery $\tau = T - t$ as

$$\sigma_F(\tau) = \sqrt{\sigma_\chi^2 e^{-2\kappa\tau} + \sigma_\xi^2}.$$

Fit a one-factor model, $\sigma e^{-a\tau}$, to the one-year and two-year contracts, which is what a desk pricing a spread between them would do. Matching their 20.7% and 20.0% needs $a = 0.035$ and $\sigma = 21.5\%$:

Months to delivery	Two-factor (market)	One-factor fit
1	39.3%	21.4%
6	24.8%	21.1%
12	20.7%	20.7%
24	20.0%	20.0%

Now price an option, expiring in six months, to receive the two-year contract and deliver the one-year, both at \$80: Example 36.5's exchange option, with both legs on the same curve. Its value is Margrabe's formula with the variance of $\ln(F_2/F_1)$ accumulated over the option's life. In the two-factor model the long-term factor moves both contracts by the same amount and drops out of the ratio, so only the short-term factor contributes:

$$v^2 = \sigma_\chi^2 \left(e^{-\kappa T_1} - e^{-\kappa T_2}\right)^2 \frac{e^{2\kappa T_0} - 1}{2\kappa},$$

with $T_0 = 0.5$, $T_1 = 1$ and $T_2 = 2$. That gives $v = 5.92\%$ and, since the two forwards are equal, a value of $80\,[2N(v/2) - 1] = \$1.89$. The one-factor model has the same formula with σ and a in place of σ_χ and κ, and gives $v = 0.50\%$ and \$0.16. Both values are forward values, undiscounted.

The one-factor model misses because of how it had to be fitted. To carry a 20% volatility out to two years it must mean-revert slowly, so the two legs have almost the same volatility; being perfectly correlated as well, they move almost in step, and the spread barely moves. The two-factor model carries the long end with ξ and leaves χ free to revert fast, and it is the fast factor that moves one contract against the other. The two legs' instantaneous correlation in the two-factor model is still 0.974 (0.927 over the option's life), and the gap from one carries all of the option's value.

And the misfit shows in the table. A desk that fitted it to the one- and six-month contracts instead would put the two-year volatility at 4.7% against 20.0%, and would overvalue the same option at \$2.91. No single σ and a fit both ends of the curve, the practical sign that a second factor is needed.

Jumps are needed for power and increasingly for gas. Example 36.3 showed what omitting them costs.

Contracts on the forward curve are modeled on the curve itself. A model that gives each

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delivery period's forward its own volatility and its correlation with the others (Chapter 32's HJM approach, applied to commodity forwards) prices block forwards and the options on them consistently with the quoted curve. Spot models with spikes and forward-curve models are therefore complements rather than rivals: the first prices what settles on a day, the second what settles on a period, and Example 36.4 shows that a spot model calibrated to daily options leaves the value of a monthly contract open by a factor of more than three.

Commodity modeling is less standardized than rates or equities. The practical discipline is to name the physical feature each model term is representing (this factor is inventory, this jump is a capacity constraint), because a commodity model with terms that correspond to nothing physical will be calibrated to fit and will fail out of sample.

36.9 BASIS: GRADE, LOCATION, AND TIMING

Section 3.3 defined basis risk generally. In commodities it is not a residual but the main event, because a hedger essentially never owns the contract grade at the contract location on the contract date, and Example 36.8 shows what that leaves behind.

Example 36.8 What the hedge leaves behind

A producer's realized price against a light sweet crude contract:

Contract (light sweet, Cushing, December)	\$78.00/bbl
Grade adjustment (heavier, higher sulfur)	−3.20
Location adjustment (wellhead to Cushing)	−2.40
Timing adjustment (November production)	−0.35
Realized price	**\$72.05/bbl**
Total basis	−\$5.95 **(7.6%)**

Call the futures price itself the **flat price**, to distinguish it from the basis. At a 32% annual volatility of the flat price and a 28% annual volatility *of the basis*:

Unhedged standard deviation (32% of \$78.00)	\$24.96/bbl
Hedged standard deviation (28% of \$5.95)	\$1.67/bbl
Risk reduction	**93.3%**

The hedge works: it removes 93% of the risk, which is why producers hedge. But the residual is real and it is different in kind. The flat-price market is liquid, continuously quoted, and hedgeable; *basis usually is none of those*. Grade differentials move with refinery configuration and with regulation on sulfur content; location differentials move with pipeline capacity and can jump when a line is constrained; timing basis moves with local storage. Three consequences follow.

Basis is hedgeable in some markets and not others. Where basis swaps trade (major US gas hubs, some crude grades), the residual can be reduced further. Where they do not, a producer's remaining exposure is to a variable with no market and no forecast.

Basis risk is worst exactly when flat-price risk is worst. A pipeline constraint that sharply widens a location differential typically coincides with the demand event that moved the flat price. This is the wrong-way structure of Section 27.9 in a physical market, and it means the 93% figure overstates the protection in precisely the episodes the hedge was bought for.

A hedger who reports the position as "hedged" has renamed the exposure rather than removed it: flat price is hedged and basis is not, and the two should be reported separately, for the reason Section 9.11 kept three valuation numbers apart.

36.10 STACK AND ROLL

The most instructive hedging failure in commodities is not a model error but a hedge that matched the exposure barrel for barrel and failed on its cash flows, and nothing in the pricing

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gives it away (Example 36.9).

Example 36.9 A hedge that looks right and still fails

A supplier sells ten years of fixed-price delivery, one million barrels a year. The far-dated futures are illiquid, so it hedges by **stacking** the whole ten-year exposure in the front month and rolling monthly.

Total commitment	10,000,000 bbl
Stacked in the front month	10,000,000 bbl
Exposure arising in year one	1,000,000 bbl
Stack ratio	**10×**

Three problems follow, and any one of them would be enough to sink the hedge.

One: the roll costs money, on the stacked notional. Section 5.11's roll yield applies ten times over:

Monthly contango	Per year	Cost per year on a $780m stack
0.5%	6.0%	$46.8m
1.0%	12.0%	$93.6m
2.0%	24.0%	$187.2m

Two: the margin timing. A price fall makes the futures lose cash immediately while the offsetting gain on the supply contract accrues over a decade. The present value discounts each year's gain from mid-year at 5%, continuously compounded:

Price fall	Futures loss (cash, now)	PV of contract gain
−10%	$ 78.0m	$ 61.4m
−20%	$156.0m	$122.7m
−30%	$234.0m	$184.1m
−40%	$312.0m	$245.5m

A hedge that is economically right and cash-flow wrong can end a firm, which is Section 8.9's margin-as-liquidity-risk in its purest form.

Three: the stack ratio is wrong even in principle. Two separate reasons apply, and they compound.

Discounting. At −10% the present value of the gain is $61.4m against a $78.0m loss, or 79%. A one-for-one stack over-hedges in present-value terms by construction, because ten years of future gains are worth less than an immediate loss of the same nominal size.

The Samuelson maturity effect adds to it. Example 36.2 showed the front contract is roughly four times as volatile as the twelve-month at $a = 1.5$. A one-for-one hedge of a ten-year exposure with front-month futures is therefore too large in volatility terms as well.

The remedy is a maturity-weighted stack (fewer front contracts, weighted by e^{-aT} and by the discount factor), and the correct hedge ratio requires the mean-reversion parameter, which cannot be estimated reliably: Example 36.7 shows that no single a fits both ends of the curve, so the value obtained depends on which end it is fitted to. So the choice is between a hedge that is knowably wrong and one that is wrong by an unknown amount, and the first is preferable only if its error is reported.

The same mismatch returned in 2022, one generation after Metallgesellschaft (Section 3.10): European generators that had sold their output forward owed variation margin in cash as gas and power prices rose several-fold, while the offsetting gain on their plants would arrive only on delivery. Example 8.9 works the arithmetic and Section 40.12 tells the episode.

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36.11 FINANCIAL PARTICIPATION AND THE CURVE

Commodity index investment (long-only, front-month, rolled mechanically) grew large enough to change the market it was invested in, by Section 22.8's mechanism. A published roll schedule creates a predictable flow: index funds sell the expiring contract and buy the next over a known window every month. A flow that is a published function of the calendar is a flow that can be anticipated, and the roll's cost to investors is the counterpart of someone else's return. The aggregate position also changed the curve's shape. Persistent long-only demand in the front months is widely argued to have contributed to periods of contango in markets that had historically been backwardated, which raised the roll cost and, in turn, made the index products perform worse than the spot returns they were sold on.

The general lesson has appeared before in this book: portfolio insurance in Chapter 18, inverse volatility products in Section 22.8, and now commodity index rolls. A strategy whose trades are a published function of an observable is a strategy whose trades other people can price. In the first two cases the cost arrived as a crash; for an index roll it shows up as underperformance rather than as a loss anyone can point to.

36.12 FREIGHT, METALS, AND OTHER MARKETS

Three markets round out the picture, and each illustrates a point the chapter has already made.

Freight derivatives settle against published route indices: forward freight agreements on the cost of moving a cargo between two ports. Freight, like power, cannot be stored: a voyage not made today cannot be banked, so the spot rate is set entirely by the balance of available ships and cargoes, and the market behaves more like power than like oil. Its economic role is to complete the locational spread of Section 36.7: an LNG cargo's destination option is only monetizable if the shipping leg can be priced, and freight is what prices it.

Base metals trade on a structure inherited from a physical warehousing system: delivery on individual days, the **prompt dates**, rather than in monthly contracts, and delivery against warrants, the title documents to metal in approved warehouses. The warehouse queue is a real economic variable: metal in a warehouse with a long load-out queue is not immediately available, so it supports the convenience yield of Section 36.1 even though the inventory statistics count it.

A short squeeze in a metal is the one place where the missing arbitrage of Section 36.1 becomes acute: a party short a physical contract cannot manufacture the metal, so the backwardation has no ceiling. The 2022 nickel episode is treated in Section 2.9 as a clearing and market-structure case; the price behavior that produced it is this chapter's asymmetry taken to its limit.

Agricultural markets add the feature that supply is annual and known only after harvest, so the curve breaks at the crop year. A calendar spread across a crop-year boundary is a different instrument from one within a year. It is the same warning as Section 36.2's, with a sharper discontinuity, because old-crop and new-crop are physically distinct goods.

The common thread is that in each market the financial contract is anchored to a physical constraint (a ship, a warehouse queue, a harvest), and the constraint, not the contract, sets the behavior the model has to reproduce.

36.13 EMISSIONS ALLOWANCES

A regulator caps emissions, issues allowances equal to the cap, and requires one surrendered per metric ton emitted; allowances trade, so *the price is the marginal cost of abatement*, the cost of the last ton of emissions cut to meet the cap. Three properties are all that matter here: they are **bankable**, so an allowance not surrendered this year can be kept for a later one; holding one is a registry entry; and borrowing one is hard, since the lending market is thin and holders

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with a compliance need will not lend. So $u = 0$ and, absent doubt about future validity, $y = 0$, and (36.1) collapses to $F = Se^{r(T-t)}$—the limiting case of a storable commodity, in which the carry relation binds tighter than for anything physical in this chapter.

Allowances are created by governments, so the primary market is an auction run by the regulator or on its behalf, and the secondary market is mostly in futures rather than in the allowances themselves. A futures contract delivers allowances into the buyer's registry account at expiration, and most volume sits in each year's December contract, which lines up with the annual compliance cycle. The main schemes are these:

Scheme	Unit	Where it trades	Main users
EU Emissions Trading System (from 2005)	EU allowance (EUA)	Auctions on the European Energy Exchange (EEX); futures and options on ICE and EEX	Power generators; steel, cement, chemicals and refining; airlines on European routes; shipping from 2024
UK Emissions Trading Scheme (from 2021)	UK allowance (UKA)	Auctions and futures on ICE	Power generators, heavy industry, airlines on UK and UK–EEA routes; domestic shipping from July 2026
California and Québec cap-and-trade	California carbon allowance (CCA)	Joint quarterly auctions; futures on ICE and Nodal Exchange	Power generators and importers, large industry, suppliers of transport fuel and natural gas
Regional Greenhouse Gas Initiative (US Northeast and Mid-Atlantic)	RGGI allowance	Quarterly auctions; futures on ICE and Nodal Exchange	Fossil-fuel power generators only
China national emissions trading scheme (from 2021)	Chinese emission allowance (CEA)	Spot only, on the Shanghai Environment and Energy Exchange	Power generators; steel, cement and aluminum added from 2025

The EU market is the deepest, and the euro prices in this section's examples are of the order of an EU allowance's price. The last column lists the compliance entities, which hold allowances because they must surrender them. In the EU, UK and North American schemes they are joined by dealers who finance compliance purchases and funds that hold allowances as a view on the cap. Importers of iron and steel, cement, aluminum, fertilizer, hydrogen and electricity into the EU owe certificates under the EU's Carbon Border Adjustment Mechanism (CBAM), priced off the allowance, for goods imported from 2026 (the first certificates are sold in February 2027), a new class of buyer with a compliance horizon rather than a financial one.

Example 36.10 A storable asset with no storage cost

Spot €72.00, financing 3.5%. With $u = y = 0$, (36.1) gives

Years	Theoretical forward
0.5	€73.27
1	€74.56
2	€77.22
3	€79.97

The one-sided bound of Section 36.1 survives, and it points the same way. A curve *steeper*

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than carry is arbitraged (buy spot, bank, sell forward), and here the arbitrage is unusually clean, because banking is costless and unlimited, so the upper bound binds tightly. A curve flatter than carry cannot be arbitraged, since that requires borrowing an allowance.

The residual is therefore the whole content. Suppose the forward trades 3% below carry at every horizon. Inverting (36.1) as in Example 36.1:

Horizon	Implied y
0.5y	6.09%
1y	3.05%
2y	1.52%
3y	1.02%

A positive implied convenience yield here cannot be physical: there is no warehouse and no queue. It prices a probability that a banked allowance will not be usable as expected, and the table shows how such a risk reads: a discount of the same size at every horizon, as a single expected loss of value would produce, appears as an implied yield decaying like $1/T$, whereas a running yield would be flat.

Two things contaminate that residual before it can be used. Dealers finance compliance inventory (an emitter buys allowances forward from a dealer holding the spot), so the curve prices a financing trade as much as an environmental scarcity and the residual is event risk *net of* a financing spread the market does not publish. And where banking stops at a phase boundary, the end of one of a scheme's multi-year trading periods, the allowance is a claim that expires, so the carry relation holds within a phase and not across one: Section 36.12's crop-year discontinuity with a legal calendar in place of a biological one, and a spread straddling the boundary is not a calendar spread.

Example 36.11 A convex supply curve meeting a vertical one

Abatement has a merit order, cheap measures first:

Cumulative abatement (Mt)	Marginal cost (€/t)
50	8
120	25
200	45
260	70
300	110
320	190

The cap fixes the supply of allowances, so a rise in required abatement cannot be met by issuing more and moves the price along this curve. Compare equal increases of 20 million metric tons (Mt), reading the price between rows by linear interpolation:

Required abatement	Clearing price	Change
190 → 210 Mt	€42.50 → €49.17	+15.7%
250 → 270 Mt	€65.83 → €80.00	+21.5%
270 → 290 Mt	€80.00 → €100.00	+25.0%
290 → 310 Mt	€100.00 → €150.00	**+50.0%**

The same 20 Mt shift moves the price 15.7% in one place and 50.0% in another, with nothing changed but the position on the curve.

Two consequences follow. Volatility rises as the cap tightens, because the curve steepens, so a maturing allowance market becomes more volatile rather than less, which is the reverse of what happens as a financial market deepens. And a value at risk (the loss that should be exceeded only

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with a small stated probability) estimated on a historical window understates the risk systematically, because the elasticity that generated the history is not the elasticity now in force. The correct state variable is the position on the abatement curve, not the price level, and a risk model that conditions on the second is fitting the wrong thing.

Two adjacent instruments are routinely priced off allowances, and each fails for a reason this book has already named. A **voluntary credit** claims a reduction against an unobservable counterfactual, so credits are not fungible and a standardized futures contract on them prices the worst deliverable. That is Chapter 6's cheapest-to-deliver problem with integrity in place of duration, and it is why most of the market stays bilateral. A **renewable certificate** carries an obligation proportional to sales rather than a cap, so supply grows with generation and Example 36.11's vertical supply does not apply; and where the obligation can be discharged at a fixed **buy-out** price whose proceeds are not returned to certificate holders (the US alternative compliance payment, not the UK scheme, which recycles them), that price is a hard cap and the certificate is a capped payoff by construction. An option on it inherits the cap, which is Section 34.12's capped cliquet in another market: past a threshold the vega changes sign with no change in the contract.

36.14 OPTIONS ON AN UNDERLYING WITH A BIMODAL DENSITY

Allowance options are quoted in Black's model like every other commodity option, and the quotes need reading differently. The reason is distributional rather than institutional, so the treatment here applies to any underlying whose uncertainty is dominated by a scheduled binary event (an election, a court ruling, a drug-trial result), and carbon is the case where such events recur.

Example 36.12 What a bimodal density does to the smile

Model the one-year allowance price as a mixture: with probability $p = 0.35$ a tightening takes the forward to €110, otherwise to €45, and within each regime the price is lognormal with 25% volatility. The mixture forward is

$$F = 0.35 \times 110 + 0.65 \times 45 = €67.75$$

and a call is priced by conditioning:

$$c(K) = e^{-rT}\Big[\, p\, c^{\mathrm{B}}(F_h, K, \sigma_w) + (1-p)\, c^{\mathrm{B}}(F_l, K, \sigma_w)\,\Big] \tag{36.5}$$

with c^{B} Black's undiscounted call, $F_h = 110$ and $F_l = 45$ the two regimes' forwards, $\sigma_w = 25\%$ the volatility within each regime, and $r = 3.5\%$, Example 36.10's financing rate. Inverted for a single Black volatility:

Strike	Call	Implied vol	Delta error
€40	28.167	45.5%	−0.036
€50	21.934	51.1%	−0.069
€60	17.418	55.0%	−0.059
€67.75	14.535	**56.4%**	−0.026
€80	10.552	56.3%	+0.034
€90	7.769	54.7%	+0.067
€100	5.474	52.6%	+0.078
€120	2.407	48.4%	+0.060

Delta error is the mixture's forward delta, with both regimes' forwards moving in proportion to F, minus Black's $N(d_1)$ at the fitted implied volatility. Three results follow.

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The implied volatility humps instead of smiling. It peaks near the money and falls from 56.4% there to 45.5% and 48.4% in the wings, the reverse of a smile and unlike an equity index's downward skew. The density has two modes and a valley between them, so an at-the-money option straddles the valley and is expensive, while the far tails, which carry only each regime's own 25% diffusion, are thinner than a lognormal's of the same variance, so the wing options are cheap. A desk that fits a standard skew parametrization to these quotes will fit a curvature of the wrong sign and will extrapolate in the wrong direction.

The at-the-money quote does not measure the total uncertainty. The mixture's standard deviation is €36.32, which a lognormal matched on variance would report as 50.3%. That is not the at-the-money implied volatility, which is 56.4%, and neither number is the volatility of a diffusion the price actually follows. Decomposing the variance shows that the regime uncertainty contributes 73% and the within-regime diffusion 27%, so nearly three-quarters of what a model calls volatility is uncertainty about which regime the price lands in, not diffusion.

The fitted volatility gives the wrong hedge. The delta error reaches −0.069 at €50 and +0.078 at €100, eight delta points on a position that the trader believes is hedged. The sign flips across the strikes, so errors at different strikes offset only by accident: the book's net error depends on its strike mix, and it accumulates a residual that looks like gamma P&L and is actually model error. The remedy is to hedge with the mixture's delta and use the implied volatility only as a quoting convention.

The right diagnostic is the density, not the volatility. Section 19.2's Breeden–Litzenberger construction recovers Figure 36.2's left panel directly from the option prices and displays the bimodality that any single volatility number averages away.

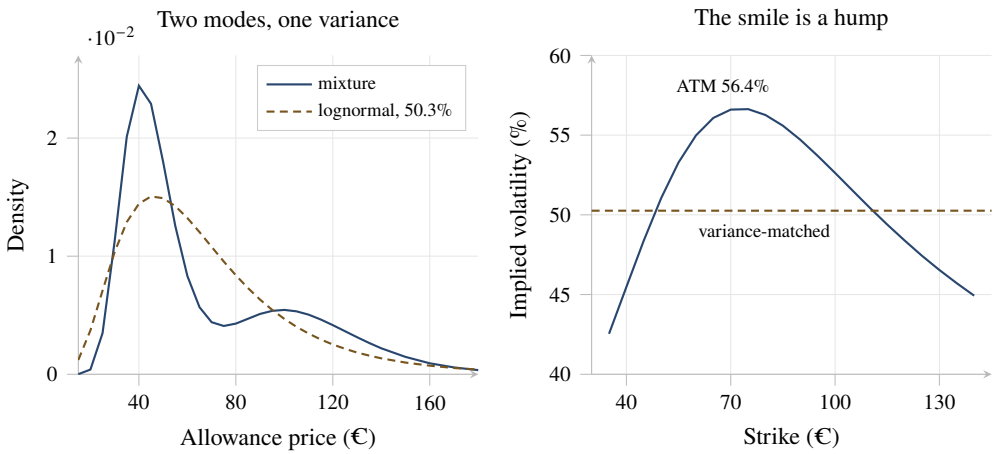


Figure 36.2: Example 36.12: mixture of lognormals, $p = 0.35$ on €110 against €45, 25% within each regime.

Figure 36.2 puts the two results side by side. The densities in the left panel have identical means and identical variances, and the lognormal's mode lies a little above the mixture's lower peak (€46 against €41) while standing only 61% as tall. From there it decays monotonically, filling the valley the mixture leaves empty (0.0095 against 0.0041 at €75) and missing the upper mode entirely. The right panel is what that misfit looks like through the only lens the market quotes in, and the hump is the diagnostic: an implied volatility surface that peaks near the money and falls in both wings is reporting a bimodal density, not a volatility.

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36.15 WEATHER DERIVATIVES

Almost every underlying so far in this book can be held, and the exceptions (the VIX of Chapter 22, the inflation index of Section 33.10) are tied by replication or by a traded bond to something that can. A weather index is tied to nothing, so replication fails entirely rather than in part.

A weather derivative settles against a measured index, most commonly **heating degree days** (HDD), or their cooling counterpart, cooling degree days (CDD), defined daily as

$$\text{HDD} = \max(65^\circ\text{F} - T_{\text{avg}}, 0), \qquad \text{CDD} = \max(T_{\text{avg}} - 65^\circ\text{F}, 0)$$

with T_{avg} the day's average temperature and a base of 18°C in Europe; the index, written HDD from here on, is the sum of the daily values over a period. The exposure it hedges is volume, not price: a gas utility sells less gas in a warm winter, and no commodity hedge addresses that, because the utility's problem is that it has less to sell, not that the price is wrong.

The index adds up a season of daily values, none of them large against the total, so it is close to normal rather than lognormal (which describes a product of growth factors, not a sum), and the Bachelier form is the appropriate one. With index mean μ, standard deviation σ, strike K, tick value A (the payment per index point) and $d = (\mu - K)/\sigma$,

$$P = A\big[\sigma\,\varphi(d) - (\mu - K)\,\Phi(-d)\big], \qquad C = A\big[\sigma\,\varphi(d) + (\mu - K)\,\Phi(d)\big] \tag{36.6}$$

with φ and Φ the standard normal density and distribution. The values are at settlement and, paid today, carry the discount factor of any cash payment. There is no volatility input in the usual sense: σ is the standard deviation of an index, estimated from history.

Example 36.13 Pricing without a hedge

November–March HDD at a reference station: mean 1200, standard deviation 140. A utility buys an HDD put struck at 1100 with a \$5,000 tick. Here $d = 100/140 = 0.7143$, and (36.6) gives

$\Pr(\text{HDD} < 1100) = \Phi(-d)$	23.8%
$E[1100 - \text{HDD} \mid \text{HDD} < 1100]$	82.2 HDD
Value of the put	**\$97,618**
Value of the matching call	\$597,618

Parity checks the pair: $C - P = A(\mu - K) = \$500{,}000$ exactly, which is the Bachelier form of put–call parity on a forward, with μ in place of a forward because the index is not an asset. Like the two values, it is stated at settlement; paid today, both sides carry the same discount factor.

That \$97,618 is an actuarial expectation, not a replication cost. No one can buy a degree day, so the writer cannot delta-hedge and the price is the expected payout plus a risk load set by capital and competition. Weather derivatives are priced like insurance and traded like derivatives.

That makes the estimate of σ, and of μ, the entire pricing problem. Two conventions compete. **Burn analysis** takes the empirical distribution of the last 20 to 40 years of the index and averages the payoff; **index modeling** fits a daily temperature process and simulates. The first is transparent and assumes stationarity; the second imposes structure and can be wrong in ways burn analysis cannot. A warming trend makes stationarity false, so detrending is a pricing input rather than a refinement: a 30-year window with a 1 HDD per year trend misstates μ by roughly 15 HDD. At this tick that is \$75,000 on a contract linear in the index, and it moves this put's value by about \$19,000, a fifth of its premium.

The geographic basis is the other half of the problem. The index is measured at one station; the utility's load is spread over a service area. For a hedge of load exposure L with the station index I, the minimum-variance ratio is $h^* = \rho\,\sigma_L/\sigma_I$ and the residual standard deviation is $\sigma_L\sqrt{1-\rho^2}$:

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Station–load ρ	Variance reduction	Residual s.d.
0.95	90%	31%
0.90	81%	44%
0.85	**72%**	**53%**
0.70	49%	71%
0.50	25%	87%

At $\rho = 0.85$ the hedge removes 72% of the variance and leaves 53% of the standard deviation, and, unlike Section 36.9's commodity basis, there is no instrument to hedge the remainder, because a contract on the utility's own service area would have no counterparty willing to price it.

The missing hedge sets the market's structure. Weather risk is diversifiable across regions and perils but not hedgeable within one, so the natural writer is a party holding many uncorrelated exposures (a reinsurer or a diversified fund), and the market's participants look accordingly more like insurers than like derivatives dealers.

36.16 CATASTROPHE BONDS

Insurance risk can also be transferred in funded form. A catastrophe bond pays a coupon and returns principal unless a defined event occurs, in which case principal is written down and paid to the sponsor (usually an insurer or reinsurer, sometimes a government or a corporation), which transfers the risk through a special-purpose vehicle that issues the bond. Structurally it is Section 25.15's funded credit protection with a physical trigger, and Part V's machinery applies to the note's mechanics unchanged.

Pricing is an expected loss plus a load, and the load is usually the larger part. The market quotes the ratio of spread to modeled expected loss as the **multiple**:

Expected loss	Spread	Multiple	Risk load
1.00%	4.50%	4.50×	3.50%
2.00%	6.50%	3.25×	4.50%
4.00%	10.00%	2.50×	6.00%
8.00%	16.00%	2.00×	8.00%

The multiple falls as the expected loss rises, which is the same pattern as credit: remote risks carry proportionally more compensation because the investor is being paid for the uncertainty in the loss estimate as well as for the loss. The loss estimate also comes from a vendor catastrophe model rather than from a market, so the multiple is partly compensation for model risk, which is why it widens after an event that the models did not anticipate, with no change in the physical hazard.

The trigger is the design decision and the trade-off is unavoidable. An **indemnity** trigger pays on the sponsor's actual losses, which is perfect for the sponsor and gives the investor exposure to the sponsor's underwriting quality. That is Section 26.9's incentive problem. A **parametric** trigger pays on a measured physical quantity, wind speed or ground acceleration at a location, which removes the incentive problem and introduces basis risk of the kind Example 36.13 quantified. A trigger that eliminates basis risk transfers the moral hazard and one that eliminates the moral hazard transfers the basis risk. Industry-loss-index and modeled-loss triggers sit between the two, each giving up some of one for less of the other; none removes both.

The diversification claim is unusually solid. Catastrophe risk is close to uncorrelated with financial markets because nothing in the markets causes an earthquake or a hurricane. That independence rests on a physical cause rather than on a correlation estimated from a sample, which distinguishes it from almost every other diversification claim in this book, where the estimate fails in the state where it is needed. The claim holds for the trigger and not for

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everything around it: in 2008 four bonds whose collateral sat behind a Lehman Brothers swap fell in price with no catastrophe, which is why such collateral is now usually held in Treasury money-market funds.

36.17 WHAT IS PRICED BY ARBITRAGE AND WHAT IS PRICED BY EXPECTATION

The instruments of the last four sections sort by how much of the book applies, and the sort matters because the formulas look identical in all three cases.

Allowance forwards: the machinery applies. Banking enforces the carry relation in one direction, and Example 36.10's residual is interpretable with the tools of Section 36.1.

Allowance options: the mechanics apply and the assumptions do not. Equation (36.5) prices them correctly and Black's formula reports the answer in units of a volatility that is not one. The hedge ratio, not the price, is where the misreading costs money: Example 36.12's 0.078.

Weather and catastrophe: almost none of it applies. There is no replication, no arbitrage bound and no hedge for the writer. These are priced actuarially, and the derivatives contribution is contract design (the index, the trigger, the settlement) rather than the valuation.

The valuation machinery of Chapter 5 and Part III derives a price from the cost of a hedge, and it returns a meaningful number only when the hedge exists. Where it does not, the same formula still returns a number, and that number is an expectation under an assumed distribution with nothing enforcing it. Both are legitimate quantities; confusing them is the error.

One practical marker separates them. If the model's output changes when you change the hedging instrument, the price is a replication cost; if it changes only when you change the distribution, it is an expectation. A desk that cannot say which of the two it is quoting does not know what its risk report means.

36.18 WHO TRADES THESE MARKETS, AND WHY

Producers hedge to protect a capital-intensive investment made years before the revenue arrives. A producer's hedge is usually a term-structure position, not a spot one, because the exposure is a production profile.

Consumers (airlines, utilities, food processors) hedge input costs, and their exposure is an average, which is why Section 33.8's Asian settlement is the norm.

Merchants and physical traders are the market's distinctive participants and have no analog in financial markets. They own the optionality of Section 36.7: storage, shipping, blending, and the right to deliver in one place rather than another. Their profit is the extrinsic value that the physical owner did not price, and it is earned by continuous re-optimization.

Utilities and generators are long the spreads of Section 36.6, whether or not they hedge them, and a generator that does not model its plant as a portfolio of options holds one anyway. In a market with an allowance price they are long the *clean* spread (power less fuel less carbon), so Section 36.6's exchange option becomes a three-asset spread requiring a correlation matrix rather than a single ρ, and the carbon leg is the one whose correlation is least stable.

Compliance buyers are the structural demand in allowances, and their hedging horizon is set by a surrender calendar rather than by a business plan, which makes their demand inelastic on known dates—an unusual and exploitable feature, and one reason some schemes split the obligation across dates (California's annual partial surrender within a multi-year period).

Utilities again, and reinsurers, are the two sides of weather. The absence of a natural opposite side within a region is why this market stayed small, and the point generalizes: a derivatives market needs not a risk but two parties with opposite exposures to it.

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Financial investors seek diversification and inflation sensitivity, and Section 36.11 describes what their participation did to the curve. In catastrophe risk their diversification claim is unusually well founded; in allowances their position is a directional bet on scarcity described as a commodity trade.

The recurring theme is that these markets are unusually close to something outside finance. The convenience yield is a fact about warehouses, the Samuelson maturity effect is a fact about how fast a shortage resolves, a negative power price is a fact about the cost of shutting down a turbine, and a carbon implied volatility is a fact about a cost curve.

36.19 WHERE THIS CAME FROM

This subject is older than the rest of the book and grew from the study of storage rather than of options. Keynes (1930) argued that the futures price sits below the expected spot because hedgers pay speculators to bear risk, the normal backwardation of Section 5.13. Working (1949) and Brennan (1958) explained the shape of the curve instead by the cost and the benefit of holding the physical commodity, which is where the convenience yield comes from.

The modern treatments make that yield stochastic, as Gibson and Schwartz (1990) did, and then ask how many factors a given commodity actually needs. Schwartz (1997) is the standard comparison and remains the natural starting point for choosing between one-, two- and three-factor specifications, because it compares them on the same data rather than advocating one.

SUMMARY

One asymmetry generates the chapter. The cash-and-carry arbitrage runs one way (buy and store, never short and un-store), so contango is capped by storage economics and backwardation is not, and the cap disappears when storage fills. The convenience yield is the residual that makes the carry relation hold, and read off the curve it measures how scarce physical inventory is now and how long the scarcity is expected to last.

Storage also sets the volatility structure. Mean reversion makes near contracts more volatile than far ones, so options on different delivery months describe different markets and cannot be compared at one volatility. Electricity, which cannot be stored at all, is the extreme case: its forward is an expectation rather than an arbitrage price, its spot price spikes, and a lognormal matched on mean and variance misprices caps in both directions, overpricing those near the money and badly underpricing those far out. Which model a power contract needs depends on what it settles on. A daily cap depends only on each day's price; a contract on a monthly average depends on how spikes cluster in time, which no model of a single day can see.

Physical assets hide options. A power plant is a strip of exchange options whose value turns on the correlation between power and gas, and high correlation makes it cheap. Storage facilities, swing contracts and tolling agreements add flexibility, valued by dynamic programming because each choice uses up capacity that a later choice would have needed. Spreads between contract months need at least two factors, since a one-factor model fitted to a realistic curve moves the months in step and prices the spread close to zero. Hedging leaves a basis residual that is worst exactly when the hedge is needed, and a short-dated hedge of a long-dated commitment can be economically right and still fail for want of cash.

An emissions allowance is the limiting storable commodity. Banking is free, so the carry relation binds tightly, and whatever residual remains measures doubt about the scheme rather than physical scarcity. The price is set where a fixed supply meets a convex abatement curve, so volatility rises as the cap tightens. Policy events make the density of a future allowance price bimodal, so implied volatility humps instead of smiling, and the Black delta errs in a direction that flips across strikes.

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At the far end the underlying cannot be held. Weather contracts and catastrophe bonds give the writer no hedge, so they are priced as an expectation plus a load for bearing the risk, and the buyer's residual exposure to its own location or portfolio cannot be removed. The distinction the chapter exists to make follows: if the output changes when you change the hedging instrument it is a replication cost, and if it changes only when you change the distribution it is an expectation.

FURTHER READING

Geman's *Commodities and Commodity Derivatives* is the standard reference, and the one that works through the physical constraints that generate the financial behavior. Clewlow and Strickland on energy derivatives is the practitioner's introduction to the modeling of Section 36.8 and is short. Schwartz and Smith's two-factor model is the paper behind that section's recommendation and is worth reading for the way it separates short-term and long-term uncertainty. On electricity specifically, Huisman and Mahieu's regime-switching model is the one behind (36.3), which is its one-day marginal; Cartea and Figueroa, and Geman and Roncoroni, add the mean reversion and jumps that Example 36.4 shows a monthly contract needs; and Weron's book surveys both families and the evidence that their fit varies from one market to the next. Eydeland and Wolyniec on energy and power risk management treats at length the spread and storage valuation of Sections 36.6 and 36.7. On the financialization question of Section 36.11, the empirical literature is contested and worth sampling from both sides; the disagreement is mostly about magnitude rather than direction. For Section 36.14, the mixture-of-lognormals treatment of event risk is developed in the literature on option prices around scheduled announcements. The carbon application is a special case of it, and reading the announcement papers first makes the hump in Figure 36.2 look inevitable rather than exotic. Weather derivatives have a small and stable literature of which the pricing material is mostly actuarial; the useful question, how to detrend, is treated mainly in the insurance journals rather than the finance ones. On catastrophe bonds the market's own offering documents set out the trigger trade-off of Section 36.16 and are worth reading in preference to summaries.

PRACTICE QUESTIONS

36.1 State the Samuelson relation and compute the twelve-month volatility at $\sigma_S = 50\%$ and $a = 1.0$. Using Example 36.2, state the one-month to twelve-month ratio at $a = 1.5$ and explain what a cross-maturity hedge ratio of one would cost.

36.2 Using Example 36.3, state the ratio of spike-model to lognormal cap values at \$200 and at \$400, and explain why calibrating the lognormal to at-the-money options does not close the gap.

36.3 Using Example 36.5, state the value of the option on one megawatt-hour at $\rho = 0.9$ and at $\rho = 0$, and explain why high power–gas correlation makes the spread option cheap rather than expensive.

36.4 Compute the 3-2-1 crack spread at crude \$80/bbl, gasoline \$2.50/gal, heating oil \$2.55/gal, and state what changes if the refinery's actual yield is 2-1-1.

36.5 Using Example 36.6, state the extrinsic share at 60% volatility and $\rho = 0.75$, and explain why the facility is usually leased by a trader rather than operated by the utility that built it.

36.6 Invert the carry relation to give the implied convenience yield, and compute it for $S = 80$, $F = 77$ at six months, $r = 4\%$, $u = 5\%$. State what a negative value would indicate and why it cannot be arbitraged away.

36.7 Using Example 36.8, state the risk reduction achieved by hedging the flat price, and explain why that figure overstates the protection in exactly the episodes the hedge was

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bought for.

36.8 Using Example 36.9, state the annual roll cost at 1% monthly contango on a \$780m stack, then state the cash loss and the present value of the offsetting gain at a 10% price fall and explain why the difference can end a firm.

36.9 Give the two independent reasons a one-for-one stack is the wrong size, state the remedy, and explain what the remedy requires that cannot be reliably estimated.

36.10 Compute the two-year allowance forward at a spot of €65 and a 4% rate, and state which side of that value is enforced by arbitrage.

36.11 Explain why a positive implied convenience yield in allowances cannot be physical, what its decay with maturity indicates, and how dealer financing of compliance inventory contaminates the reading.

36.12 Using Example 36.11, state the price change from two different 20 Mt shifts in required abatement, and explain what that implies for a value-at-risk estimated on a historical window.

36.13 Explain how a buy-out price caps a renewable certificate and what that does to the vega of an option written on it.

36.14 Using Example 36.12, state the at-the-money implied volatility and the two wing values, explain the shape, and state what a standard skew parametrization fitted to these quotes would get wrong.

36.15 Explain why the variance-matched lognormal volatility of 50.3% differs from the 56.4% at-the-money implied volatility, and what each measures.

36.16 Using Example 36.12, state the largest delta error, explain why a book spread across strikes does not reliably net it out, and state what the residual will be mistaken for.

36.17 Explain why the Bachelier form rather than Black's is appropriate for a degree-day index, and use (36.6) to verify the put–call parity relation in Example 36.13.

36.18 Explain the difference between burn analysis and index modeling, state what each assumes, and quantify the effect of a 1 HDD-per-year trend on Example 36.13's contract.

36.19 Using Example 36.13, state the minimum-variance hedge ratio as a formula in the two standard deviations, state the residual standard deviation at $\rho = 0.85$, and explain why the remainder cannot be hedged.

36.20 Explain why a catastrophe bond multiple falls as the expected loss rises, and why multiples widen after an unanticipated event with no change in the physical hazard.

FURTHER QUESTIONS

36.21 Example 36.2 gives the volatility term structure under mean reversion. Derive it from a mean-reverting spot process, then determine what the term structure looks like when storage is abundant, and use the result to explain why the observed term structure steepens when inventories fall.

36.22 Example 36.5 shows a 133% valuation error from ignoring correlation. Derive the spread option's sensitivity to ρ, determine where in the heat-rate spectrum the error is largest, and explain why a fleet of plants with different heat rates cannot be valued with a single correlation assumption.

36.23 Example 36.6 decomposes storage into intrinsic and extrinsic value. Formulate the storage operator's problem as a dynamic program with injection and withdrawal rate constraints,

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state how the constraints reduce the extrinsic value, and explain why a fast-cycling facility is worth more per unit of capacity than a slow one.

36.24 Example 36.8 finds a 93.3% risk reduction with a residual that is worst when the hedge is most needed. Model the basis as correlated with the flat price under stress, recompute the risk reduction conditional on a large flat-price move, and state how a producer should size the hedge given that conditional figure.

36.25 Example 36.10 reads the deviation from carry as a price of event risk and observes that the implied yield it produces decays as $1/T$. Formalize this: model the allowance as paying off only if the scheme survives to T, derive the implied survival probability from the curve, state what else could produce the same discount, and propose a test on the term structure that would distinguish the explanations.

36.26 Example 36.12 produces a delta error that changes sign across strikes. Derive the mixture's delta in closed form from (36.5), determine the strike at which the error is largest as a function of p, and construct the portfolio of vanilla options that would hedge the regime exposure directly rather than through delta.

36.27 Example 36.13 finds 53% residual standard deviation at a 0.85 correlation. Determine how many independent station contracts would be needed to bring the residual below 20%, state the correlation structure your answer assumes, and explain why the market has not developed such multi-station products.

36.28 Example 36.4 holds each day's price distribution fixed and varies only how spikes cluster. Prove that the strip of daily caps cannot depend on the clustering, derive the variance of the monthly average as a function of the episode length, and explain why options on block futures trade at implied volatilities far below those of daily options in the same market.

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Chapter 37
Crypto Derivatives

Chapter 36 worked through markets where the valuation machinery of Chapter 5 and Part III fails, and in each case it failed because of the underlying: electricity cannot be stored, an emissions allowance has a supply a legislature sets, a degree-day index cannot be held at all. This chapter's underlying breaks none of those rules. A bitcoin is storable at essentially no cost, perfectly fungible, divisible to eight decimals, pays no dividend, has no convenience yield, no seasonality, and no delivery constraint. By the standards of Chapter 36 it is the best-behaved underlying in the part—better behaved than crude oil, which has to be put somewhere, and far better behaved than an allowance whose validity is a political question.

So Chapter 5's carry relation applies without modification. With no income and no storage cost, the arbitrage-free futures price is $F = Se^{rT}$, the cash-and-carry arbitrage can be run in both directions, and the basis trade of Chapter 6 transfers across with the algebra unchanged. When market prices depart from that relation, as Section 37.1 shows they do by several percentage points a year, the reason is not the underlying.

What is missing is everything around the contract. There is no central counterparty, no segregation of client collateral, no settlement cycle that pauses trading (the offshore venues mark and liquidate continuously, with no pause in which to collect margin or manage a default), no enforceable netting opinion of the kind Chapter 27 called worth more than any risk model, and no lender of last resort. The pricing of Chapter 5 and Part III assumes an infrastructure it never names, and this is the market where that assumption is visible because it is absent.

The distinction between the contract and the infrastructure around it organizes the chapter, and it is not a technicality. The largest custody failure in the market's history was not caused by a mispricing, a model failure, or a move in the underlying, and Section 37.10 argues that describing it any other way misplaces the lesson.

Sections 37.1 to 37.5 take the contracts in turn: listed futures and their basis, the effect of a staking yield, perpetual futures, options, and the inverse contract with its choice of numeraire. Section 37.6 looks at the producer and Section 37.7 at the spot ETFs that narrowed the gap to the regulated market; Sections 37.8 and 37.9 are the decentralized venue and the settlement asset. Section 37.10 is the custody failure, and Section 37.11 says who trades these derivatives and why.

37.1 LISTED FUTURES AND THE BASIS

Regulated futures on bitcoin have traded since 2017, and the offshore venues list futures on hundreds of tokens. The pricing is Chapter 5's, unchanged.

Example 37.1 The basis trade, transplanted

Spot is \$60,000 and an offshore venue's three-month futures price \$61,800. Crypto markets quote the basis the other way round from Section 3.3's spot minus futures, and as a percentage: the futures premium over spot, $(F - S)/S$. Here that is $1{,}800/60{,}000 = 3.00\%$ over the quarter, or **12.55% annualized** with quarterly compounding ($1.03^4 - 1$), against a dollar financing cost of 5%

a year on the same compounding basis. In Section 5.9's terms that 5% is the whole cost of carry, since a coin pays no income and costs nothing to store, while the futures contract is priced as if the cost of carry were 12.55%. Buying spot, selling the futures contract and waiting therefore earns the gap: **7.55%** a year over the cost of carrying the position.

Traders rarely put up the full position. Leverage here is the position's notional divided by the capital posted against it (for one unit of a futures contract, the futures price divided by the margin on it), so 20% initial margin is 5× leverage, and the return on that capital is roughly the 7.55% multiplied by the leverage (taking the coin itself as financed in full at the 5%):

Initial margin	Leverage	Return on capital
50%	2.0×	15.1%
35%	2.86×	21.6%
20%	5.0×	37.8%

This is Section 6.5's Treasury basis trade with the numbers moved up by two orders of magnitude, and the reason the excess over financing is 7.55% rather than a few basis points is not that the arbitrage is harder. It is that three things Chapter 6 took for granted are absent: there is no CCP, so the position is an unsecured exposure to a venue; margin is set at the venue's discretion rather than by a published risk model, and can be raised without notice; and the collateral is held by the counterparty, so the trade's return is conditional on the counterparty surviving the quarter.

The basis is therefore not an inefficiency but a price. This is Section 5.15's point that arbitrage needs a balance sheet, here one entrusted to the venue.

The return in Example 37.1 does not show the trade's main risk, which is a shortage of cash rather than a loss of value, as Example 37.2 shows.

Example 37.2 A hedged trade that a rally can close

The basis trade has no net exposure to the price: the spot leg and the short futures position offset, so the profit is locked in at entry. The cash flows are not flat. The futures position is marked to market as the price moves; the spot is not (unless the coin itself is posted as the margin, as an offshore venue allows). So a rally calls cash on one leg against an unrealized gain on the other.

At $60,000 spot the futures price is $61,800, and 20% initial margin on the contract is $12,360 per unit. The cash called is the variation margin the short futures position pays as the futures price rises; with the basis held at 3%, a rally raises the futures price by the same percentage of $61,800:

Rally	Cash called	Share of margin	Profit at expiration
+10%	$6,180	50%	unchanged
+20%	$12,360	100%	unchanged
+33%	$20,394	165%	unchanged
+50%	$30,900	250%	unchanged

A 20% rally, the margin rate itself, exhausts the margin on a position whose profit at expiration has not changed. The trade is liquidated for want of cash while the hedge is working exactly as intended, and the liquidation forfeits the 7.55%, which was locked in only for a position held to expiration. The futures loss is realized in cash at once, while the offsetting gain on the coin exists only as a mark until the coin is sold; if it is sold alongside, it recovers nearly all of that loss but none of the carry; if it is kept, it is an unhedged long.

This is Section 8.9's margin call as a liquidity risk, and the reader has met it twice already: it is Example 36.9's stack and roll, the problem that brought down Metallgesellschaft's oil hedge, and it is Chapter 8's central point, that a hedge can be correct in value and still fail on timing. What this market adds is that the venue sets the margin, so the schedule in the table is not a parameter of the trade but a decision someone else may revise mid-position.

Hence the leverage is not free. At 5× the headline return is 37.8%, and the rally that ends the

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trade is 20%. At 70% volatility, a stressed quarter for bitcoin (Example 37.7 below), a coin with no expected drift in its price rises 20% at some point within a quarter with a probability of about 55% (the touch probability of Section 34.3, at zero drift). Held this way, with the coin outside the margin account, the trade is more likely than not to be liquidated before the futures contract expires and the 7.55% is earned.

The direction of the basis is informative in the same way Chapter 36's curve was. A persistent premium says leveraged long demand exceeds the capital willing to finance it, and the premium collapses, occasionally inverting, when that demand unwinds. It shows how traders are positioned, measured by a market price rather than a survey, and it is easier to read than Chapter 36's convenience yield, which is never quoted and has to be backed out of the curve as a residual. Here the premium itself is quoted, and because a coin pays no income and costs nothing to store, subtracting the dollar interest rate leaves the excess over carry of Example 37.1 with nothing else to estimate.

37.2 WHEN THE UNDERLYING PAYS A YIELD

This chapter opened by saying the underlying pays nothing. That is true of bitcoin and false of ether, the token of the Ethereum network and the second-largest asset in the market, and the correction is worth making because it moves the carry relation from Chapter 5's case of an asset that pays nothing to its case of an asset with a known yield.

A **proof-of-stake** asset pays its holder for committing coin to validating the network. That is a continuous yield q on the asset itself, so the forward price is Chapter 5's known-yield case (Example 37.3 prices it):

$$F = Se^{(r-q)T}. \tag{37.1}$$

The yield q is not set by anyone's discretion. On Ethereum the protocol determines it, and it has two parts. The larger is newly issued coin paid to validators, at a rate that falls as more coin is staked: approximately $166.3/\sqrt{M}$ a year when M ether in total is staked, or about 2.5% at the roughly 44 million staked in late 2026. The more coin is committed, the thinner the reward is spread, so the yield regulates how much of the supply is staked. The smaller part is the transaction fees and the payments for ordering transactions that a validator collects when it proposes a block; these vary with network activity and typically add a few tenths of a percent. A staking provider keeps a commission, typically 10% to 25%, out of both.

Two consequences follow for (37.1). First, q is not known in advance, since both the amount staked and the network's activity move; a forward embeds the yield expected over its life, not a contractual rate. Second, the issuance part is paid in new coin, so it is partly a transfer from holders who do not stake, whose share of the supply shrinks. Unlike a dividend, it is not paid out of anyone's earnings, and a holder who does not stake is diluted, not merely forgoing income.

Example 37.3 A dividend the asset did not previously have

Spot \$3,000, a 5% dollar rate and the staking yield q, both continuously compounded, and three months, so the forward is $3{,}000\,e^{(0.05-q)\times 0.25}$:

Staking yield q	Forward	Quarterly basis	Annualized
0.0%	\$3,037.74	+1.258%	+5.13%
2.5%	\$3,018.81	+0.627%	+2.53%
3.5%	\$3,011.27	+0.376%	+1.51%
5.0%	\$3,000.00	0.000%	0.00%
7.0%	\$2,985.04	−0.499%	−1.98%

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The basis is computed as in Example 37.1: the premium $(F - S)/S$ over the quarter, annualized by compounding four quarters. At $q = r$ the futures price equals spot, and above it the curve inverts. The inversion says nothing about scarcity. The asset now pays a yield, and a persistent backwardation that would signal scarcity in Chapter 36 signals here only that the staking yield exceeds the dollar financing rate.

So Example 37.1's basis is not a clean positioning indicator for a staked asset. The observed premium mixes leveraged demand with the yield, and a desk reading it as sentiment will misread it whenever q moves. The two are separable only if the staking yield is quoted independently, which it is, so the correction is available to anyone who makes it.

The first US spot ether ETFs of Section 37.7 were launched without staking, and that has a cost. Grayscale's ether funds began staking in October 2025, and BlackRock's staked trust followed in March 2026, staking 70% to 95% of its ether and keeping the rest unstaked to meet redemptions. A fund holding coin without staking it forgoes the yield: on a $10 billion fund at about 2.5%, that is $250 million a year, and because most of the yield is new issuance, the fund's holders are diluted by it rather than merely missing it. In exchange it buys liquidity.

Staked coin is not immediately deliverable. Exiting a validating position means joining a queue (exit and withdrawal together usually take under a week, but took over 40 days in September 2025, when many validators exited at once), so the deliverable supply is smaller than the outstanding supply. A fund that must meet redemptions daily cannot hold all its assets behind any queue, which is why a staked fund keeps part of its coin unstaked, and the cash-and-carry arbitrage of Section 37.1 needs unstaked coin to close.

This qualifies the chapter's opening claim in a specific way. The underlying is storable at no cost *and* it now offers a yield for making it temporarily undeliverable, so a holder faces a genuine choice between earning q and retaining delivery capacity. That is Chapter 36's convenience yield reappearing—not as an inference from a curve, but as a posted rate a holder gives up to keep the asset liquid.

37.3 PERPETUAL FUTURES

A **perpetual futures** contract (a **perpetual** for short) has no maturity. It never delivers and never expires, so nothing forces convergence to spot. What replaces convergence is a payment: at fixed intervals, usually every eight hours (some venues hourly), longs pay shorts a **funding rate** set by the gap between the perpetual's price and a spot index, plus, on most venues, a small fixed interest component (0.01% per eight hours); when the perpetual trades far enough below the index the payment reverses.

The mechanism inverts the usual construction. A dated futures contract is held to spot by the *prospect* of delivery, and the cost of carry is embedded in the forward price. A perpetual is held to spot by an explicit cash flow instead, and that cash flow takes the place of the premium in a dated contract. A long in a dated contract pays the carry through the price: it buys at a premium over spot, and the premium shrinks to nothing by delivery. A perpetual's price never converges, so there is no premium to shrink, and without a payment a long perpetual would be leveraged exposure to the coin with nothing paid for the financing. The funding rate is that payment. It is not a cost of carry in Chapter 5's sense, a quantity fixed by interest rates and storage, but a price set by the demand for leverage. While the perpetual trades at a premium, the funding rate has to cover at least what an arbitrageur who sells the perpetual and holds the coin pays to hold it, and on top of that it carries whatever premium leveraged longs pay, just as the dated basis of Example 37.1 does. Example 37.4 converts one into the other.

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Example 37.4 Funding is the basis, quoted differently

Funding is paid every eight hours, so there are $3 \times 365 = 1{,}095$ periods a year:

Funding per 8 hours	Annualized (simple)	Equivalent 3-month premium
+0.0100%	+10.95%	+2.74%
+0.0300%	+32.85%	+8.21%
+0.0500%	+54.75%	+13.69%
−0.0200%	−21.90%	−5.48%

A quoted funding rate of 0.01% per eight hours and a three-month futures contract at a 2.74% premium are the same statement about the price of a levered long, and a desk running both should see them agree. When they do not, the difference is the cost of the two venues' credit and margin terms rather than anything about bitcoin. The first row is also the venues' default: inside a narrow band around the index the formula pays the fixed 0.01% whatever the gap, so funding reads demand only once it moves off that default.

Chapter 5 had to *infer* a convenience yield as the residual that made an equation hold, and Chapter 36 spent a section extracting one from a curve. The corresponding quantity here, the price of holding a levered long, is printed every eight hours rather than backed out of a curve.

The perpetual exists because a contract that never settles asks less of the infrastructure than a dated one: there is no expiration to settle and nothing for its holders to roll. The instrument is an adaptation to missing infrastructure, which is the chapter's theme in a single product. The adaptation has since come onshore: on May 29, 2026 the CFTC approved a cleared bitcoin perpetual for a US exchange. Inside a CCP the contract keeps its funding rate; what changes is everything around it.

Liquidation replaces the margin call. There is no cure period, and no one at a broker deciding whether to give the client more time. When margin falls below a threshold the venue closes the position at once. In traditional markets the same trigger usually produces a margin call instead (from the broker on exchange-traded futures, from the counterparty under a credit support annex on OTC derivatives, from the broker on a securities margin account), and a margin call starts a clock: the client has about a day to raise cash, and can see whether the move reverses before committing it. The difference lies with the venue, not the coin, since bitcoin futures on CME carry margin calls like any other futures contract, and have kept them since CME moved its crypto futures to round-the-clock trading on May 29, 2026. On an offshore venue the trigger ends the position. The client can add collateral at any moment up to that point, but must do so before knowing whether it will be needed, so protection means collateral posted early and left idle, or a position watched around the clock—and a fast market can cross the threshold before anyone can act. Section 8.9's collateral loop runs here in minutes rather than days. And because leveraged positions use a few standard leverage multiples, their liquidation levels cluster and the forced flow is a known function of the price: Section 35.7's barrier clustering in another market, with one difference that Example 37.5 draws out.

Example 37.5 A cascade with published barriers

Take \$12 billion of leveraged long perpetual positions: \$10.8 billion in the four tiers below and \$1.2 billion at lower leverage. A position at ℓ times leverage has margin worth $1/\ell$ of its notional, so it liquidates on a fall of roughly $1/\ell$ (a little less in practice, since the venue closes it before the margin is entirely gone):

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Leverage	Liquidates on	Notional in tier	Cumulative forced selling
25×	−4%	$1.2bn	$1.2bn
20×	−5%	$1.8bn	$3.0bn
10×	−10%	$4.2bn	$7.2bn
5×	−20%	$3.6bn	$10.8bn

The table assumes that every position's leverage is measured from the current price and that no one adds margin as the price falls. A 10% decline mechanically forces $7.2 billion of selling, which is 60% of those positions, and the selling arrives into a market already falling. The second-round effect is the point: that flow pushes the price further, which triggers the next tier. It is the sale at Section 35.6's knock-in barrier repeated at every tier, without the buying that precedes it there, and no issuer decides when to trade: the venue's automated liquidation system does.

But the disclosure is the opposite of Section 35.7's. Autocallable barriers cluster because issuers independently choose round fractions, and no issuer can see the aggregate; that was the whole difficulty. Here open interest and the venues' leverage schedules are public in real time, and data firms publish the liquidation profile estimated from them, so anyone can see its shape.

That does not make the cascade smaller. It changes who gains from the cascade: the flow is visible in advance to the traders best placed to trade ahead of it, and the resulting anticipation is itself part of why the moves are abrupt. Transparency relocated the problem rather than removing it, which is worth setting beside Chapter 2's case for uniform, transparent margin.

The largest recorded case is October 10, 2025, when, amid new US tariff announcements aimed at China, trackers such as CoinGlass counted about $19 billion of liquidations in roughly a day, and at least one large venue closed profitable positions by auto-deleveraging (Section 37.10).

37.4 OPTIONS

Options trade on the offshore venues, on CME futures since 2020 and, since 2024, on the regulated ETFs of Section 37.7. The pricing is Section 16.5's. The offshore contracts are European and cash-settled, and each is priced against the futures contract of the same expiration, so the model is Black's, with the futures price in place of spot; market makers hedge with that contract and with the perpetual. That choice matters more here than anywhere in Part III. Black's formula needs only the forward, so it takes in the basis of Section 37.1 and the staking yield of Section 37.2 without anyone having to estimate them, which is Section 16.8's rule of getting the forward right. A value computed from spot and the dollar rate assumes a forward of Se^{rT}, missing the part of the basis that financing does not explain, and so misprices the option, as Example 37.6 shows. Time is counted in calendar days over 365, since the market never closes.

Example 37.6 Pricing off the forward the market trades

Take spot at $60,000, the three-month futures price of $61,800 from Example 37.1, a strike of $60,000, 70% volatility and a 5% dollar rate, continuously compounded:

Forward used	Call	Put
Futures price, $61,800 (Black)	$9,273	$7,496
Se^{rT}, $60,755	$8,661	$7,916

Pricing from spot and the dollar rate undervalues the call by $612, or 7%, and overvalues the put by $420, or 6%. The two errors are the same mistake: put–call parity, $c - p = e^{-rT}(F - K)$, ties both prices to the forward, and the second row uses the wrong one. The hedge is in the same

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contract. The call's delta against the futures contract is $e^{-rT}N(d_1) = 0.595$, with

$$d_1 = \frac{\ln(61{,}800/60{,}000) + (0.70^2/2) \times 0.25}{0.70\sqrt{0.25}} = 0.2595,$$

so a dealer who sells the call buys 0.595 futures contracts per coin.

These options differ from an equity index's in three ways. The first is the level and the term structure. At-the-money implied volatility on bitcoin mostly ran at 35% to 55% in 2026, spiking above 80% in February (ether and smaller tokens higher), and short-dated volatility jumps above long-dated in a sharp rally as well as a sharp fall, where equity indices invert mainly in a sell-off (Section 19.6).

The skew changes sign. Equity index skew has pointed one way since 1987; crypto skew reverses with the market, with calls in demand in a rally and puts in demand in a decline. This is the pattern Section 19.1 found in currencies, whose tilt also changes sign as sentiment moves, rather than the one it found in equity indices, and for a similar structural reason: there is no leverage effect (the tendency of a stock's volatility to rise as its price falls, because a falling share price raises the firm's debt relative to its equity) and no large, persistent buyer of protection of the kind equity portfolios supply, so the skew reflects whichever side is currently buying options.

The volatility risk premium is large and unstable. Section 19.10's two to four volatility points on equity indices is often ten points or more here, which is why so much of the retail product in this market is a covered call in some form. What that level of volatility costs is worth seeing in currency rather than in percentage points.

Example 37.7 What a 70% volatility costs

Consider a thirty-day at-the-money straddle on a \$60,000 spot:

Volatility	Straddle	Share of spot	Break-even move
15%	\$2,059	3.43%	±3.43%
30%	\$4,116	6.86%	±6.86%
50%	\$6,857	11.43%	±11.43%
70%	\$9,591	**15.99%**	±15.99%
90%	\$12,318	20.53%	±20.53%

An equity index at 15% costs 3.43% of spot to straddle for a month; this market at a stressed 70% costs 15.99%. The buyer needs a 16% move in thirty days merely to return the premium, and the seller collects 16% of spot for a month of risk, which is why the yield products of Section 37.11 can advertise returns that look impossible and are not.

The premium is real and it is not a riskless profit. To first order a delta-hedged short option earns, per year,

$$\frac{1}{2}\,\Gamma S^2 \left(\sigma^2_{\text{implied}} - \sigma^2_{\text{realized}}\right),$$

so per unit of dollar gamma ΓS^2 it earns $\frac{1}{2}(\sigma^2_{\text{implied}} - \sigma^2_{\text{realized}})$, which for the first row of the table below is $\frac{1}{2}(0.70^2 - 0.60^2) = 6.5\%$:

Implied	Realized	Profit per unit of ΓS^2
70%	60%	+6.5% a year
70%	55%	+9.4% a year
70%	75%	−3.6% a year

Five points of adverse surprise cost 3.6% a year, so the strategy's sign depends on a quantity

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no one observes in advance, which is Section 22.1's point about delta-hedged options, with larger numbers.

37.5 INVERSE CONTRACTS AND THE CHOICE OF NUMERAIRE

A large part of this market trades in contracts that settle in the coin rather than in dollars, and the convention is not cosmetic. An **inverse** contract has a notional fixed in dollars, margin posted in the coin, and settlement paid in the coin: a long position of notional Q entered at F_0 pays

$$\Pi_{\text{coin}} = Q\left(\frac{1}{F_0} - \frac{1}{S_T}\right). \tag{37.2}$$

This is Section 28.6's numeraire question posed by a contract specification rather than by a model. Convert (37.2) at the terminal price and the dollar payoff is $Q(S_T/F_0 - 1)$—*exactly linear*. Leave it in coin and it is not, as Example 37.8 shows.

Example 37.8 Linear in one currency, curved in the other

Take a long inverse position, \$100,000 notional, entered at \$60,000:

Price	Move	P&L in coin	P&L in dollars
\$30,000	−50%	−1.6667	−\$50,000
\$42,000	−30%	−0.7143	−\$30,000
\$48,000	−20%	−0.4167	−\$20,000
\$72,000	+20%	+0.2778	+\$20,000
\$90,000	+50%	+0.5556	+\$50,000
\$120,000	+100%	+0.8333	+\$100,000

The dollar column is a straight line and the coin column is not. A 50% fall costs **1.6667 coin** while a 50% rise earns only 0.5556, and the coin gain is capped at $Q/F_0 = 1.6667$ however far the price rises while the coin loss is unbounded. A long inverse position is short gamma measured in coin, with $\partial^2\Pi_{\text{coin}}/\partial S^2 = -2Q/S^3$.

With an initial margin of 0.5 coin, the long's equity is $0.5 + Q(1/F_0 - 1/S)$ coin, which moves as follows:

Price	Equity (coin)	Equity (dollars)
\$54,000	+0.3148	\$17,000
\$48,000	+0.0833	\$4,000
\$42,000	−0.2143	−\$9,000

At entry the 0.5 coin is \$30,000, so a dollar-margined trader at this leverage is wiped out by a 30% fall; in coin the equity is gone where $0.5+Q(1/F_0-1/S) = 0$, at $S = 1/(1/60{,}000+0.5/100{,}000) =$ \$46,154, a 23% fall. Equity falls faster than a dollar-margined trader's (\$17,000 against \$20,000 at \$54,000), because the margin is itself coin and loses value with the price, so liquidation arrives earlier than a dollar-margined trader would compute from the same leverage. The collateral and the exposure are the same asset, which is Chapter 27's wrong-way risk in its purest form, written into the contract rather than arising from a correlation.

The convention has a practical origin. An inverse contract needs no dollar leg, so a venue can run it without touching the banking system: the whole contract lives in the asset it references. That mattered when venues had no bank accounts, and it is why the oldest contracts in this market are inverse ones; the newer contracts margined in dollars or in dollar tokens (stablecoins, Section 37.9) now carry most of the volume. It also suits a miner, which can sell one against its coin and fix the coin's dollar value without ever holding a dollar.

The pricing follows Chapter 28. Let $F = Se^{rT}$ be the carry forward of Chapter 5, the

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frictionless market's price, setting aside the venue credit that Section 37.1 found in the basis; because the discounted coin price is a martingale under the dollar risk-neutral measure, F is also $E[S_T]$, the expectation under that measure. In dollars the payoff $Q(S_T/F_0 - 1)$ is linear and worth

$$e^{-rT} Q\left(\frac{E[S_T]}{F_0} - 1\right) = e^{-rT} Q\left(\frac{F}{F_0} - 1\right),$$

which is zero at $F_0 = F$, so the fair entry price is the ordinary forward. With the coin as numeraire, a value in coin is the coin-measure expectation of the payoff in coin, and the change of measure has density $(S_T/S)\, e^{-rT} = S_T/F$, so

$$E^S\left[\frac{1}{S_T}\right] = E\left[\frac{S_T}{F} \times \frac{1}{S_T}\right] = \frac{1}{F},$$

and the contract is worth

$$Q\left(\frac{1}{F_0} - \frac{1}{F}\right) \text{ coin},$$

zero at the same price. Mixing the two measures gives a wrong answer. For a lognormal price with volatility σ, $\ln S_T$ has mean $\ln F - \sigma^2 T/2$ and variance $\sigma^2 T$ under the dollar measure, so averaging the coin payoff's $1/S_T$ with dollar-measure probabilities gives

$$E\left[\frac{1}{S_T}\right] = e^{-\ln F + \sigma^2 T/2 + \sigma^2 T/2} = \frac{e^{\sigma^2 T}}{F},$$

not $1/F$. The extra factor $e^{\sigma^2 T}$ is Section 30.6's quanto term, with $1/S_T$ as the asset and S_T as the exchange rate, so their correlation is -1, and no correct valuation of this contract contains it.

37.6 THE PRODUCER: MINING AS A SPREAD POSITION

This market has a producer, as Chapter 36's power market does. A miner spends electricity and capital to receive coins, so it is long the coin and short power: Section 36.6's spark spread with a different output. The hedging problem lies where the comparison fails (Example 37.9).

Example 37.9 A miner's break-even, and the hedge the protocol supplies

A miner produces 500 coins a year against \$14 million of electricity and \$4 million of other costs. Break-even is \$36,000 a coin, and because the costs are fixed while revenue moves with the price, the profit moves much more than the price does:

Coin price	Revenue	Costs	Profit	Profit/revenue
\$30,000	\$15.0m	\$18m	−\$3.0m	−20.0%
\$45,000	\$22.5m	\$18m	+\$4.5m	+20.0%
\$60,000	\$30.0m	\$18m	+\$12.0m	+40.0%
\$90,000	\$45.0m	\$18m	+\$27.0m	+60.0%

The output is not a fixed quantity, and that is what separates this from the spark spread. The network issues new coins at a fixed rate, a batch with each block of transactions, and a miner wins blocks in proportion to its share of the **hash rate**, the computing power all miners together apply to the puzzle each block requires. The protocol sets how hard that puzzle is, and its **difficulty adjustment** resets it at intervals so that blocks keep arriving on schedule however much hardware is running. If a price fall drives competitors off, the survivor's share rises at once but blocks slow down, so its output rises only when the difficulty falls and blocks return to schedule.

Suppose the price falls from \$60,000 to \$42,000 and 20% of the network's hash rate leaves:

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	Revenue	Change
Before	\$30.0m	—
Before the difficulty adjustment	\$21.0m	−30%
After the difficulty adjustment	\$26.3m	**−12.5**%

The protocol converts a 30% revenue shock into a 12.5% one. The miner is long the price, and it is also long the exit of competitors, which happens when the price falls; the two offset, so the exposure is to price × share, and share moves against price.

Three qualifications keep this from being a free hedge:

- the difficulty adjustment comes at intervals of roughly two weeks, so the offset arrives after the cash-flow problem it would have solved;
- the protection comes from the miners who leave, and those are the high-cost ones, so a high-cost miner is near the exit itself when it needs protection most, while the low-cost miner, which needs it least, gains the most; and
- the hedge works through quantity, so a forward sale of a fixed amount fits neither direction: after a fall the miner produces more than it sold and the surplus is unhedged, and after a rise it produces less and must buy coin at the high price to deliver.

This explains how miners that hedge do so. They sell a fraction of expected output forward rather than all of it, and favor puts over forwards, because a forward that must be delivered into is a short position in a market where the quantity is itself uncertain. This is volume risk. The contract built for it is the swing contract of Section 36.5, in which the buyer chooses how much to take within limits; a miner would want the mirror image, with the choice of how much to deliver on its own side. The nearest listed contract is a hashprice futures contract, which fixes revenue per unit of hash rate rather than per coin, and it trades thinly, so most miners approximate the hedge with a strip of listed puts and accept the mismatch.

Power contracting is the other half of the position and it is where the interesting structures are. A miner with an interruptible power contract has sold an option to the grid: it agrees to shut down on demand in exchange for a lower price. Interruptible contracts are old (aluminum smelters have signed them for decades), but a miner is an unusually good counterparty for one, because what it gives up by shutting down is coin: the shutdown cost is observable at every instant as the coin price times the forgone production, less the power not bought, so the option's exercise value can be read from a quoted price. Several grids now contract with miners this way.

37.7 SPOT ETFS AND THE NARROWING OF THE GAP

The most consequential development in this market was not a derivative. United States spot bitcoin exchange-traded funds began trading in January 2024, spot ether funds in July 2024, and options on the bitcoin funds in November 2024.

What that changed is precisely the thing this chapter says is missing. An ETF share sits in ordinary custody, its options clear through the Options Clearing Corporation, the positions net under an enforceable framework, and the collateral is segregated. None of it altered bitcoin itself; it altered the infrastructure wrapped around it.

The gap between the two venues is therefore measurable, and it is the price of that infrastructure. An option on the ETF and an option on the offshore venue reference nearly the same underlying and differ in counterparty, segregation, netting, and margin treatment. Net of the contract differences (the ETF options are American, settle in fund shares and trade only in US hours) and of any difference in liquidity, the spread between their implied volatilities is Chapter 27's capital and Chapter 8's collateral, expressed in volatility points, and it is the cleanest available estimate of what the missing infrastructure was worth.

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37.8 AUTOMATED MARKET MAKERS

An automated market maker, the commonest design of decentralized exchange, has no order book. It holds an inventory of two assets and quotes by a rule: the **constant-product** rule keeps

$$x\,y = L^2,$$

where x is the inventory of a token, y the inventory of a dollar stablecoin, counted here as dollars, and L is set by what liquidity providers have deposited: a trade moves x and y along the curve and, apart from the fee it leaves in the pool, leaves their product, and so L, unchanged. $L = \sqrt{xy}$, the geometric mean of the two inventories, measures the pool's size and is called its **liquidity**; a deeper pool moves its price less for a trade of a given size. A stablecoin is a token its issuer promises to redeem for one dollar, backed by a reserve of dollar assets, and is the settlement asset of Section 37.9. Anyone may trade against the pool at the price the rule implies. That price is the ratio of the inventories, $S = y/x$, because along the curve $xy = L^2$ a small trade exchanges dx tokens for $dy = -(y/x)\,dx$ dollars. A buyer of the token pays stablecoin in and takes tokens out, so y rises, x falls and the price y/x goes up.

Providers of that inventory are told they are earning fees. They are, and Example 37.10 shows what they are paying for them.

Example 37.10 Providing liquidity is selling gamma

At a price S the two conditions $xy = L^2$ and $y/x = S$ pin the inventory to $x = L/\sqrt{S}$ and $y = L\sqrt{S}$, so the position is worth

$$V(S) \;=\; x\,S + y \;=\; 2L\sqrt{S}. \tag{37.3}$$

Two derivatives tell the whole story. The delta is $V'(S) = L/\sqrt{S} = x$, exactly the token inventory, so the pool's exposure to the price is always its current holding, and the rule rebalances that holding for it: it sells the token as the price rises and buys as it falls. And

$$V''(S) \;=\; -\frac{L}{2\,S^{3/2}} \;<\; 0,$$

so the position is concave in the price everywhere. A liquidity provider is short gamma, and Chapter 18 established what a short gamma position earns and what it loses.

Against simply holding the initial basket the pool falls short, and the market calls the shortfall **impermanent loss** (IL), a name suggesting the loss will reverse, which it does only if the price returns to where it started. The shortfall depends only on the ratio $k = S/S_0$ of the current price to the price S_0 at deposit. The inventory deposited, $x_0 = L/\sqrt{S_0}$ and $y_0 = L\sqrt{S_0}$, would be worth $x_0 S + y_0 = L\sqrt{S_0}\,(1+k)$ if simply held, while by (37.3) the pool is worth $2L\sqrt{S} = 2L\sqrt{S_0}\,\sqrt{k}$. Impermanent loss is measured as the pool's value relative to the held basket's, $\mathrm{IL} = V(S)/(x_0 S + y_0) - 1$:

$$\mathrm{IL}(k) \;=\; \frac{2L\sqrt{S_0}\,\sqrt{k}}{L\sqrt{S_0}\,(1+k)} - 1 \;=\; \frac{2\sqrt{k}}{1+k} - 1$$

Price ratio	0.25×	0.5×	0.8×	1.25×	2×	4×
Shortfall	−20.00%	−5.72%	−0.62%	−0.62%	−5.72%	−20.00%

It is symmetric in k and $1/k$: the shortfall depends on the size of the move in log terms, not on its direction, because neither token is privileged. And a payoff concave in the underlying is a short position in options, here a written straddle.

The cost of being short gamma, the expected rate at which the pool falls behind the basket when the price has no drift, which the market calls its bleed, has a simple closed form. Chapter 18's gamma cost per unit time is

$$\frac{1}{2}\,|V''|\,\sigma^2 S^2 = \frac{1}{2}\,\frac{L}{2S^{3/2}}\,\sigma^2 S^2 = \frac{L\sigma^2\sqrt{S}}{4},$$

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and dividing by the position's value $2L\sqrt{S}$:

$$\text{cost per year} = \frac{\sigma^2}{8} \quad \text{of the position, whatever the price and whatever the pool size.} \tag{37.4}$$

Volatility	40%	60%	80%	100%	120%
Fee yield needed to break even	2.00%	4.50%	**8.00%**	12.50%	18.00%

The expected value of the shortfall also has an exact closed form for a driftless lognormal price, $e^{-\sigma^2 T/8} - 1$, because the bleed is charged on the pool's current value and so compounds: at 80% volatility the exact loss over three months is 1.980% against $\sigma^2 T/8 = 2.000\%$, and over a year 7.688% against 8.000%. Multiplying the rate by the horizon gives only the first term of the exponential's expansion, so it always overstates the loss: by 4% of the exact figure at one year, but at 120% volatility over three years, where $\sigma^2 T = 4.32$, its 54.0% overstates the exact 41.7% by 29%.

So the question a liquidity provider should ask is the one an option seller asks: is the fee income above or below the cost of the volatility I am short? At 80% volatility the pool must return 8% a year in fees before the provider breaks even, and a pool advertising a 6% fee yield on a pair that volatile is advertising a loss.

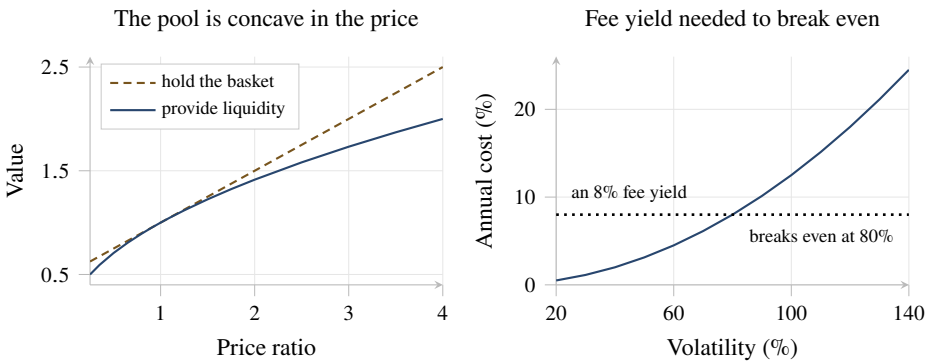


Figure 37.1: Left: pool value against holding the same initial basket, normalized to 1. Right: (37.4).

Figure 37.1 is the reason this section is in a derivatives textbook rather than a technology one. The left panel is a written straddle drawn in unfamiliar coordinates, and the concavity is the product, not a defect of it. A provider who understood the position as a short option would ask about the premium; one who understands it as "earning yield" will not.

The concentrated-liquidity variant makes the option clearer. Providers may confine their inventory to a price range, which concentrates the gamma and raises both the fee income and the bleed—and outside the range the position is entirely one asset and the delta is constant. That is a strip of options written at strikes inside the range, with fees that accrue only while the price stays in it, like Section 33.6's range accrual; it is sold to retail users with no options vocabulary at all.

37.9 THE SETTLEMENT ASSET

Most trades on the offshore venues settle in a **stablecoin** rather than in dollars, and that changes what the quoted prices mean. A perpetual quoted at 60,000 is quoted in a token that its issuer promises to redeem at a dollar, so the futures price carries the issuer's credit whether or not anyone prices it that way.

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The peg is not a law of the market; it is an arbitrage with a cost. If the token trades below a dollar, someone buys it and redeems it at par for the profit. That trade works only while redemption works, which requires the reserve to be liquid, the issuer to be operating, and the redemption window to be open. Remove any of the three and the token stops being a dollar and becomes a claim on a portfolio, valued as Part V values any other claim.

Example 37.11 What the discount was pricing

In March 2023 Circle disclosed that $3.3 billion of the roughly $40 billion reserve behind its USDC stablecoin, or 8.25%, sat at a bank that had just failed. Take the rest of the reserve as sound and price the token as a claim:

Recovery on the exposed reserve	Fair value	Discount
100%	1.0000	0.00%
50%	0.9588	4.13%
0%	0.9175	8.25%

The token traded to about 0.8774, a 12.3% discount, against an arithmetic worst case of 8.25%. No recovery assumption produces that price, so the extra 4.0 points were not pricing the disclosed exposure at all.

They were pricing the redemption mechanism itself. The arbitrage that holds the peg needs an open window; a holder who fears the window closes before their turn will sell at any price above what they expect a queue to yield. So the discount decomposes into a credit loss on a known exposure and an option-like charge for access to the redemption facility, and only the first is in the disclosure.

The link to the rest of the book is direct. The failure of Silicon Valley Bank (Section 40.14) reached the derivatives market of this chapter through the settlement asset—not through any position, model, or counterparty in it. A trader perfectly hedged in coin terms lost 12.3% on the units the hedge was denominated in, which is Section 7.8's currency basis argument with the funding currency issued by a private firm.

Two consequences follow for anyone marking a book in this market. First, a stablecoin-denominated profit and loss is not a dollar profit and loss, and the two diverge exactly when the market is stressed—which is when the distinction matters. Second, margin posted in a stablecoin is collateral with credit risk and correlation to the very events that trigger margin calls, so Chapter 27's wrong-way risk applies to the collateral rather than to the exposure. A haircut on the settlement asset is the obvious remedy, and where venues apply one it is a percent or two, because a larger one would concede that the unit of account is not a dollar. Regulation is now supplying part of what the haircut did not. The US GENIUS Act of 2025, whose rules take effect on January 18, 2027, requires one-for-one reserves in cash, deposits and short-dated Treasuries, and redemption at par, and CFTC staff letters since December 2025 let futures commission merchants accept payment stablecoins, including those of national trust banks, as customer margin.

37.10 THE VENUE IS THE COUNTERPARTY

The failure of FTX in November 2022 is usually told as a story about crypto. It is more useful as a story about Chapter 2, because every layer that chapter described was missing and the absences can be listed one by one.

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Case Study 37.1 A default waterfall with no layers

Chapter 2 set out the sequence in which a clearing house absorbs a member default. Set that sequence beside what a large crypto venue had in place of each layer:

Chapter 2's waterfall	The venue
Defaulter's initial margin	Held by the venue, unsegregated
Defaulter's default fund contribution	None
The CCP's own capital, ahead of members	No distinction between the venue and its operator
Surviving members' mutualized fund	An "insurance fund" sized and used at the venue's discretion
Assessments and recovery tools	Losses socialized by unilateral decision

Bitcoin settled every block throughout. No contract mispriced, no model failed, and the underlying did what it always does. Client assets were not segregated, so a customer's exposure was not only to the market but to the venue, and the venue was under the same control as the trading firm on the other side of its customers' trades.

Section 27.2 called a jurisdiction's netting opinion (a lawyer's opinion that close-out netting will be enforced if a counterparty fails) worth more to a dealer than any risk model. This market shows what that means by its absence. There was no netting opinion, no bankruptcy-remote structure, and no supervisor that required either, so the entire apparatus of Part V (exposure at default, close-out netting, the margin period of risk) had nothing to attach to.

The lesson is not that the instruments were exotic. They were plain instruments: futures, perpetuals, and spot balances. What was exotic was the absence of the infrastructure every earlier chapter assumed.

Once the liquidated trader's margin and the venue's insurance fund are gone, a crypto venue uses **auto-deleveraging**, whose counterpart Part I's clearing houses keep as a last resort. When a liquidation cannot be filled at a price that leaves the venue whole, the shortfall is met by closing profitable positions on the other side, ranked by their profit and leverage. A trader who is right about the market and fully margined can have the position taken away to fund somebody else's loss, which is Chapter 2's tear-up brought forward to the layer after the insurance fund, without the mutualization, the disclosure, or the governance.

37.11 WHO TRADES CRYPTO DERIVATIVES, AND WHY

Leveraged directional traders dominate trading and are the perpetual's main users: a contract with no expiration, no delivery, and high leverage is the cheapest way to express a view. Their liquidation levels are the forced flow of Section 37.3, which makes their aggregate position a market-structure variable rather than a private one.

Basis traders and market-neutral funds harvest the basis of Example 37.1, and they are the capital that keeps the futures anchored to spot. They are also the first to leave, since their return is a spread and their risk is the venue, so the basis widens when the venue is in doubt.

Miners are the natural commercial hedgers, with a revenue stream in coin and costs in ordinary currency, chiefly for electricity, which is the position of Chapter 36's generator. Those that hedge the coin revenue do so by selling forward and buying puts.

Companies and funds holding coin sell covered calls for yield, giving up the price rise above the strike for a premium. It is the same trade, with the same asymmetry, as the yield products of Chapter 35.

Liquidity providers supply the inventory of Section 37.8, mostly without pricing the option

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they have written.

The market's participant mix is unusual in one respect. In most markets this book describes, the natural hedger came first and the speculator followed. Here the speculative flow built the market and the commercial hedgers arrived afterward, which is why the instruments are optimized for leverage rather than for delivery, and why the infrastructure lagged the volume by a decade.

SUMMARY

The underlying is the easy part and the infrastructure is the hard part. A bitcoin is storable at no cost, fungible, divisible, and pays nothing, so Chapter 5's $F = Se^{rT}$ is the arbitrage-free price. What is absent is the CCP, the segregation, the settlement pause, the netting opinion, and the lender of last resort: the infrastructure Chapter 5 and Part III assume without naming it.

That explains the prices. A futures basis far wider than the cost of financing is the venue's credit being priced, not an inefficiency. On a staked asset the basis also mixes in the yield, so it no longer reads as positioning alone. Options are priced off the forward the market trades rather than off spot. The perpetual is the only genuinely new contract: with nothing forcing convergence, a funding rate paid every eight hours does delivery's work and prints a price that Chapter 5's markets leave to be inferred. Liquidation replaces the margin call, and because open interest and leverage schedules are public, the forced flow can be estimated in advance, which relocates Section 35.7's barrier clustering rather than removing it.

Two conventions carry more risk than they appear to. An inverse contract is linear in dollars and curved in the coin, and its margin is coin, so liquidation arrives earlier than dollar leverage suggests. And most offshore trades settle in a stablecoin, so the quoted price carries an issuer's credit and the value of access to redemption. In both cases collateral and exposure are the same risk: Chapter 27's wrong-way risk sitting in the contract and in the unit of account.

The producer and the liquidity provider carry options they do not price as such. A miner holds Section 36.6's spark spread, long the coin and short power; the difficulty adjustment is an automatic hedge that arrives late and works through quantity, which is why miners sell only part of their output forward and lean on puts, and a miner on an interruptible power contract has sold the grid an option to shut it down. Providing liquidity to a constant-product pool is a written straddle that bleeds $\sigma^2/8$ a year whatever the price, so only substantial fees let the provider break even at this market's volatilities.

FTX is a story about Chapter 2. Every waterfall layer was missing, yet bitcoin settled every block throughout: no contract mispriced and no model failed. Client assets were not segregated, so the exposure was to the venue as well as to the market. The regulated route of the 2024 ETFs supplies the missing infrastructure. The instruments were never the problem.

FURTHER READING

The primary sources in this market divide sharply: venue documentation of funding-rate and liquidation mechanics shows how the contracts actually behave and is worth reading directly; most of the valuation literature applies Chapters 5 and 15 to the new contracts. On the perpetual contract, He, Manela, Ross and von Wachter and Ackerer, Hugonnier and Jermann derive the price the funding rule implies; they and the venues' specification documents are the place to start, before the secondary literature. On automated market makers, Milionis, Moallemi, Roughgarden and Zhang on loss-versus-rebalancing derive the $\sigma^2/8$ rate of Section 37.8 from the Black–Scholes setting, and the connection to Chapter 18's gamma cost is immediate. The FTX bankruptcy filings are the account of what Section 37.10 describes, and they enumerate what was and was not held. On the regulated products, the fund prospectuses set out the custody and creation mechanics that Section 37.7 argues are the whole of the change.

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PRACTICE QUESTIONS

37.1 Using Example 37.1, state the annualized basis and the return on capital at 20% initial margin, and give three reasons the spread is wider than Chapter 6's Treasury basis.

37.2 Using Example 37.2, state the rally that exhausts the posted margin on a fully hedged basis trade, and explain why a position that has lost nothing can be closed.

37.3 Explain why the leverage in Example 37.1 and the rally in Example 37.2 must be quoted together, and which earlier chapter's case has the same shape.

37.4 Using Example 37.3, state the forward price and basis at a 3.5% staking yield, and the yield at which the futures price equals spot.

37.5 Explain why a persistent backwardation in a staked asset does not mean what the same shape means in Chapter 36, and what must be quoted to separate the two.

37.6 Explain what holds a perpetual futures contract to spot in the absence of delivery, and why the mechanism is an adaptation to missing infrastructure.

37.7 Using Example 37.4, convert a funding rate of 0.03% per eight hours into an annualized rate and into the equivalent three-month futures premium.

37.8 Explain why liquidation on a crypto venue is not a margin call, and what the client gives up when it replaces one.

37.9 Using Example 37.5, state the forced selling triggered by a 10% decline, and explain why publishing the liquidation profile does not make the cascade smaller.

37.10 Explain why crypto skew changes sign with the market while equity index skew does not, and name the chapter whose pattern it resembles.

37.11 Using Example 37.8, state the coin profit and loss of a long inverse position at a 50% fall and at a 50% rise, and explain why the two are not symmetric.

37.12 Explain why an inverse contract's valuation is a change of numeraire, and what convexity term a desk introduces by averaging the coin payoff under the dollar measure.

37.13 Using Example 37.9, state the break-even coin price, and the revenue change from a 30% price fall before and after the difficulty adjustment.

37.14 Explain why the difficulty adjustment is only a partial hedge, giving the three qualifications, and why a hedging miner prefers puts to forwards.

37.15 Using Example 37.11, compute the fair value at 50% recovery on the exposed reserve, and explain what the observed 12.3% discount priced that the disclosure did not.

37.16 Explain why margin posted in a stablecoin is wrong-way risk in the collateral, and what remedy Section 37.9 says venues apply, where they apply it at all, at only a percent or two.

37.17 Using Example 37.10, state the shortfall at a price ratio of 2 and of 4, and explain why the function is symmetric in k and $1/k$.

37.18 Derive the $\sigma^2/8$ bleed of (37.4) from Chapter 18's gamma cost, and explain why it does not depend on the pool size.

37.19 Explain why the FTX failure is described as a custody failure rather than a market failure, and what the underlying did throughout.

37.20 Explain what auto-deleveraging is, how it differs from its analog in Part I, and why a correctly positioned and fully margined trader can still lose the position.

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FURTHER QUESTIONS

37.21 Example 37.4 treats the funding rate and the futures basis as two quotations of one price. Construct the arbitrage that enforces the relation, determine what frictions would let the two persist at different levels, and estimate how large a gap those frictions could sustain.

37.22 Example 37.5 treats the liquidation profile as static. Model it dynamically, so that the selling from one tier moves the price into the next, determine the condition on the leverage distribution under which the cascade is self-limiting rather than self-sustaining, and state what a venue could change to enforce that condition.

37.23 Example 37.9 shows that the difficulty adjustment offsets part of a price fall with a lag. Formulate the miner's revenue as a product of price and share, model the share as a delayed response to price, and determine the forward-sale fraction that minimizes the variance of revenue.

37.24 Example 37.11 decomposes a discount into a credit loss and a charge for redemption access. Build a model of the second component as an option on the window's remaining capacity, calibrate it to the observed 12.3%, and state what the calibrated parameter would have to mean.

37.25 Example 37.10 values a pool under geometric Brownian motion. Redo the analysis under a jump-diffusion, determine whether the $\sigma^2/8$ result survives, and state what a provider should charge for the jump component that the diffusion formula omits.

37.26 Concentrated liquidity confines the inventory to a price range. Formulate the resulting position as a strip of written options with the fee stream as a range accrual, derive the fee yield needed to break even as a function of the range width, and determine the width that maximizes expected profit at a given volatility.

37.27 Section 37.7 argues the ETF-to-offshore volatility spread prices market infrastructure. Design the measurement: state which contracts to compare, what must be controlled for, and how you would separate the infrastructure premium from a pure liquidity difference.

37.28 The snapshot in Section 37.10 maps a crypto venue onto Chapter 2's waterfall. Design a waterfall that would work for a venue with no members in Chapter 2's sense, state who funds each layer, and identify which layer is hardest to fund and why.

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